COMPLETE DIPLOMA ENGINEERING HANDBOOK

DC Machines & Traction Motors

A complete learning and examination handbook covering DC-machine construction, generators, motors, starters, testing, speed control and electric-traction control—with derivations, diagrams, characteristic curves, solved numericals, Malayalam support, practice and full model answers.

COURSE CODE
3032

PROGRAMME
Diploma in Electrical & Electronics Engineering

SEMESTER
3

CREDITS
3

CATEGORY
Program Core

PERIODS
3/week • 45/semester

Start Learning

PDF note
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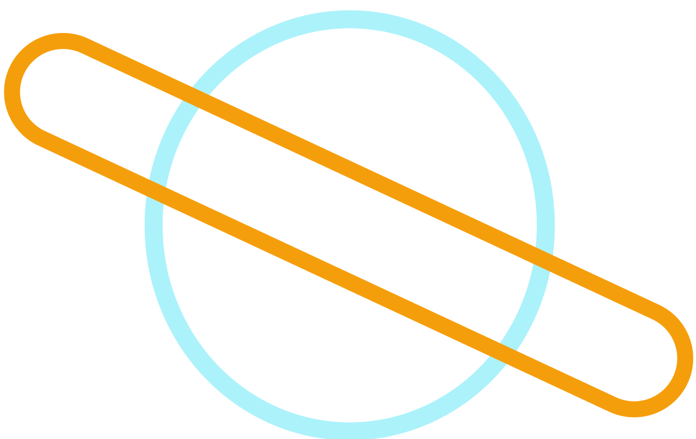

Four-module pathway

Module 1
Construction, generator action, EMF, losses, efficiency and windings

Module 2
Armature reaction, commutation, characteristics and parallel operation

Module 3
Motor action, torque, characteristics, starters and testing

Module 4
Speed control, traction motors, transitions and electrical braking

COURSE OVERVIEW

Why DC machines remain essential

DC machines convert energy in both directions. As generators they convert mechanical input into direct-current electrical output. As motors they convert DC electrical input into mechanical torque. Their transparent magnetic and electrical relationships make them ideal for learning electromechanical energy conversion, while traction applications demonstrate high starting torque, controlled acceleration and electrical braking.

Energy conversion

Faraday's law explains generated EMF; force on a current-carrying conductor explains motor torque. The same physical machine can operate in either direction.

Industrial relevance

DC motors are used where wide speed control, high starting torque or predictable torque-current relationships are required.

Traction relevance

Series motors historically suit locomotives, trams and cranes because their torque is very high at low speed and heavy current.

Typical applications

Generators

Battery charging, laboratory supplies, exciters, electroplating and welding.

Shunt motors

Lathes, fans, blowers, pumps and machine tools requiring nearly constant speed.

Series motors

Traction, cranes, hoists and elevators requiring high starting torque.

Compound motors

Presses, conveyors and rolling mills with changing load and better speed regulation.

HOW TO USE THIS HANDBOOK

Learn, trace, calculate and practise

1. Understand
Read the physical principle before memorising any equation.

2. Trace
Follow magnetic flux, current, power and motion arrows in every diagram.

3. Calculate
Use the solved format: given data, formula, substitution, unit and interpretation.

4. Verify
Attempt module questions, model paper and answer-key checks without looking first.

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OFFICIAL COURSE INFORMATION

Objectives and prerequisite

Official course objectives

1. To familiarize students with the construction and operation of DC machines.
2. To recognize the classification and applications of DC machines.
3. To analyze the performance of DC generators and DC motors.
4. To provide basic knowledge of electric traction systems.

Prerequisite

Topic: Magnetism and Electromagnetism

Course: Elementary Concepts of Electrical Systems

Semester: 2

How the prerequisite supports this course

Magnetic circuit

Flux, reluctance and permeability explain the path of field flux through yoke, poles, air gap and armature.

Electromagnetic induction

Faraday’s law and Lenz’s law explain generated EMF and back EMF.

Force production

A current-carrying conductor in a magnetic field experiences force, producing motor torque.

Fleming’s rules

Right-hand rule gives generator direction; left-hand rule gives motor-force direction.

OFFICIAL OUTCOMES

Course outcomes and CO-PO mapping

CO	Description	Duration	Cognitive level
CO1	Identify the construction and operation of DC generator.	10 hours	Applying
CO2	Identify the electrical characteristics and uses of DC generators.	10 hours	Applying
CO3	Select a DC motor for specific applications based on performance characteristics.	12 hours	Applying
CO4	Choose various methods of speed control of DC motors and traction motors.	11 hours	Applying
—	Series Tests	2 hours	—

CO-PO mapping

Course outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	Interpretation
CO1	3	—	—	—	—	—	—	Strongly mapped to PO1
CO2	3	—	—	—	—	—	—	Strongly mapped to PO1
CO3	3	—	—	—	—	—	—	Strongly mapped to PO1
CO4	3	—	—	—	—	—	—	Strongly mapped to PO1

Mapping scale

3 – Strongly mapped; 2 – Moderately mapped; 1 – Weakly mapped.

MODULE-WISE LEARNING ROADMAP

Official outcomes, hours and cognitive levels

Module outcome	Official description	Hours	Level
M1.01	Illustrate the constructional details of DC machines.	3	Understanding
M1.02	Explain the principle of operation of a DC generator.	2	Understanding
M1.03	Derive EMF equation of DC generator and solve problems.	3	Applying
M1.04	Compare the armature windings of a DC machine.	2	Understanding
M2.01	Illustrate armature reaction and its effects in DC generators.	2	Understanding
M2.02	Explain commutation process in DC generator and methods for improving commutation.	2	Understanding
M2.03	Illustrate the characteristics of various types of DC generators and solve problems.	3	Applying
M2.04	Outline parallel operation and uses of various DC generators.	3	Understanding
Series Test I		1	—
M3.01	Explain the working principle of DC motor and solve various motor parameters.	4	Applying
M3.02	Illustrate the performance characteristics of DC motors and select motors for specific applications.	4	Applying
M3.03	Illustrate the working of various types of starters.	2	Understanding
M3.04	Outline the procedures for testing of DC motors.	2	Understanding
M4.01	Identify different types of speed control of DC motors.	3	Applying
M4.02	Summarize the use of DC series motors in electric traction.	2	Understanding
M4.03	Illustrate speed control of traction motors.	3	Understanding
M4.04	Explain braking of traction motors.	3	Understanding
Series Test II		1	—

MODULE 1 • CO1 • 10 HOURS

Construction and Operation of DC Generator

M1.01–M1.04: construction, generator principle, EMF equation, losses, efficiency and armature windings.

Electrical-machine idea

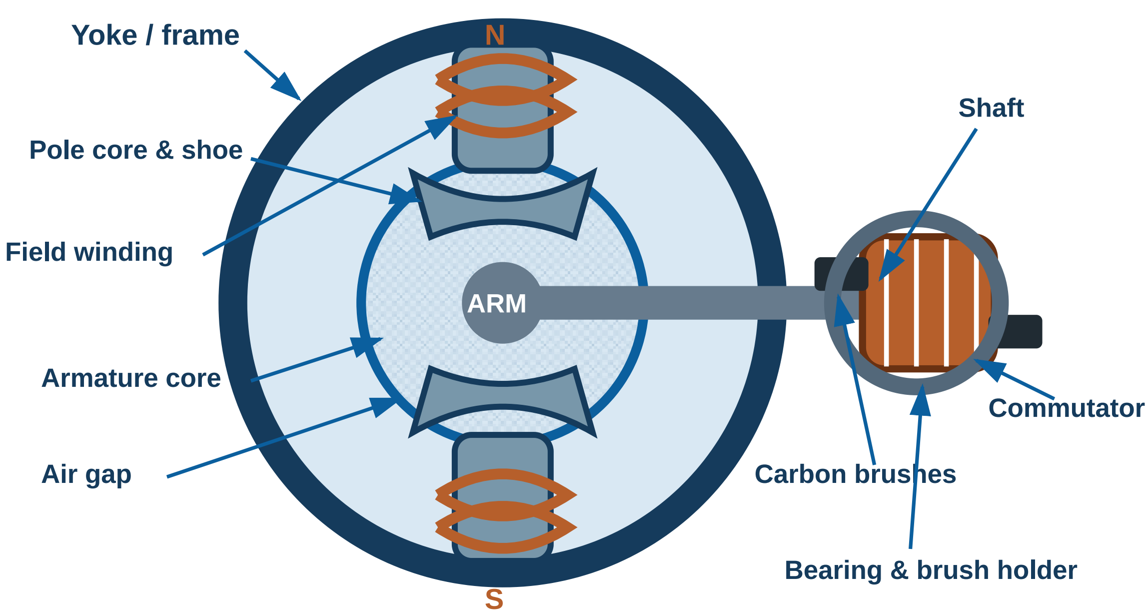
An electrical machine links an electrical system and a mechanical system through a magnetic field. In generator operation, a prime mover rotates conductors and electrical power is obtained. In motor operation, electrical current produces torque and mechanical output.

Reversibility

The physical DC machine is reversible. The same armature, field system, commutator and brushes can operate as a generator or a motor; only the direction of energy flow changes.

1. Constructional details of a DC machine

The stationary field system creates the main magnetic flux. The rotating armature carries conductors in which EMF is induced or current flows. The commutator and brushes form the mechanical switching system that provides DC at the terminals and unidirectional torque.



The yoke completes the magnetic path and supports the poles. Laminated armature steel reduces eddy-current loss. The commutator and brushes connect rotating conductors to the external DC circuit.

Part	Main purpose	Typical material / construction
Yoke	Mechanical support and low-reluctance return path for flux	Cast iron for small machines; rolled/cast steel for larger machines
Pole core	Carries field winding and conducts flux	Laminated or solid steel depending on size
Pole shoe	Spreads flux over armature, reduces reluctance, supports coil	Laminated steel
Field winding	Produces main magnetomotive force	Insulated copper conductor
Armature core	Provides slots and magnetic path	Thin silicon-steel laminations keyed to shaft
Armature winding	Carries generated/current-carrying conductors	Insulated copper coils placed in slots
Commutator	Mechanical rectifier/current reverser	Hard-drawn copper segments insulated by mica
Brushes	Sliding electrical contact with commutator	Carbon/graphite or copper-graphite
Shaft & bearings	Transmit torque and support rotation	Steel shaft with ball/roller bearings
Interpoles	Improve commutation by neutralising reactance voltage	Small poles between main poles, series with armature
Compensating winding	Neutralises armature reaction under pole faces	Conductors embedded in pole shoes, series with armature

Why laminations are used

Alternating flux in armature teeth and core induces eddy currents. Dividing the core into insulated thin laminations greatly increases the resistance of unwanted current paths and reduces eddy-current heating. Silicon steel also reduces hysteresis loss.

Cooling and mechanical support

Ventilation ducts, fans and open frames remove heat. End brackets, bearings, shaft and yoke preserve a uniform air gap. Poor alignment creates noise, vibration, brush sparking and unequal magnetic pull.

2. Magnetic circuit and generator principle

Main flux leaves the north pole, crosses the air gap, enters the armature, divides through the armature core, crosses the opposite air gap into the south pole and returns through the yoke. When armature conductors cut this flux, an EMF is induced according to Faraday's law.

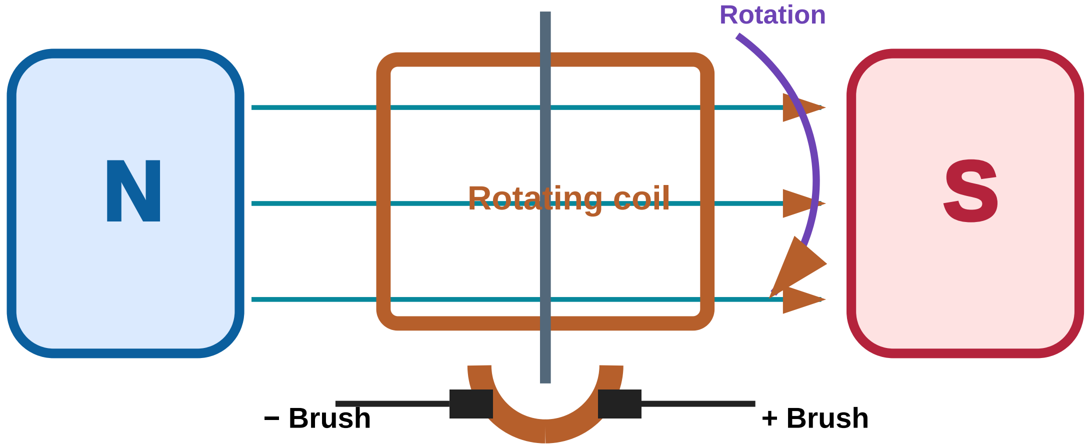
$$e = -N_c \, d\Phi/dt$$

The negative sign represents Lenz's law: induced EMF opposes the cause producing it.

Generator action

Mechanical rotation causes conductors to cut magnetic flux. Fleming’s right-hand rule gives the induced-current direction: first finger = field, thumb = motion, middle finger = induced current.

Malayalam Support: □□□□□□□□□ □□□□□□□□□ □□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□□□□ EMF □□□□□□□□□□□□□. □□□□□ □□ □□□□□□ □□□□□□□□□□□□ induced current-□□□□ □□□ □□□□□□□□□□□.



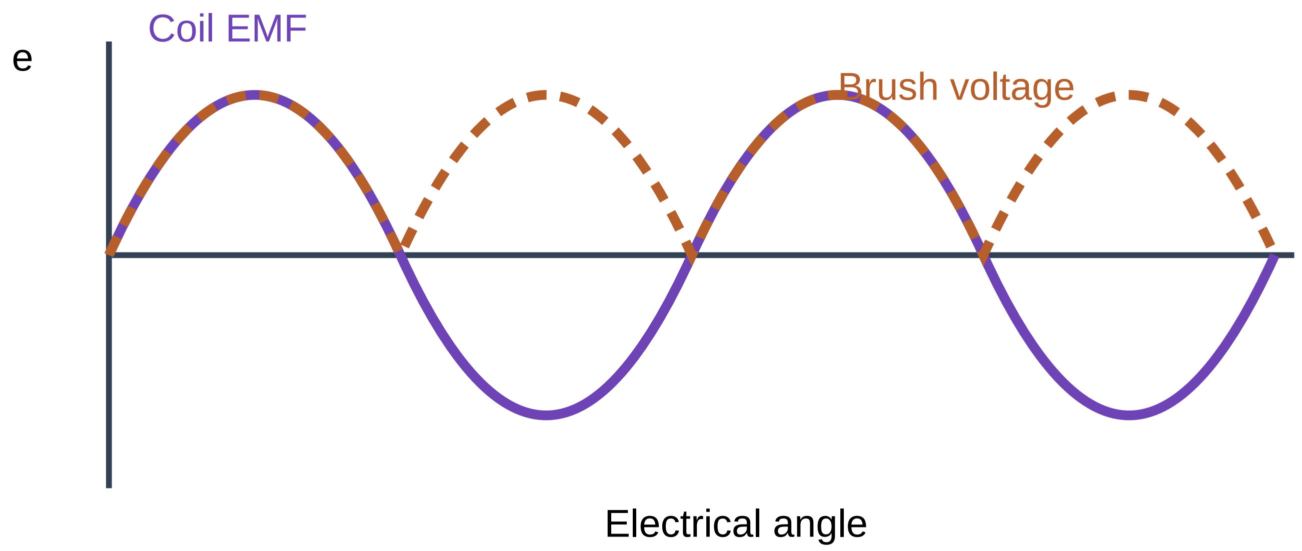
The EMF in each coil side reverses every half-turn, so the armature EMF is alternating. The split-ring commutator reverses coil connections at the same instant, keeping brush polarity unchanged.

3. Simple-loop waveform and commutator rectification

Coil positions during one revolution

Angle	Flux-cutting rate	Coil EMF	Brush output
0°	Zero	0	0
90°	Maximum	+Emax	+Emax
180°	Zero	0	0
270°	Maximum opposite	−Emax	+Emax
360°	Zero	0	0

Generated and rectified waveform



4. Classification by field excitation

Separately excited

Field winding receives current from an independent DC supply. Flux can be controlled independently of armature load.

Shunt

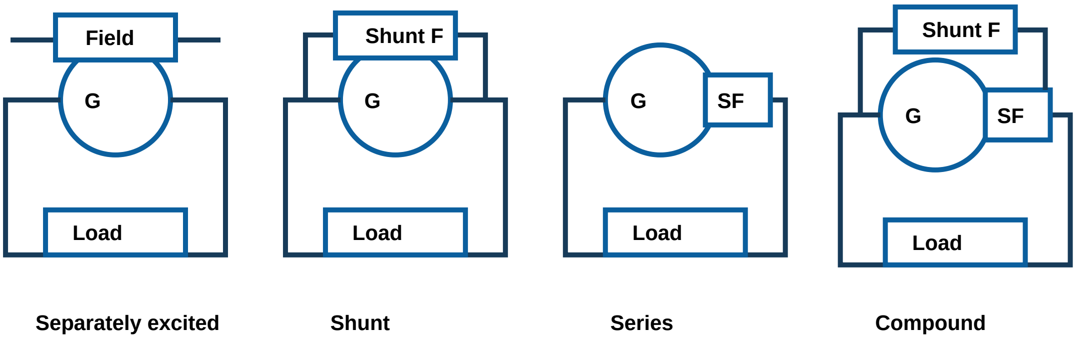
Field winding is connected across armature terminals. It has many turns, high resistance and small field current.

Series

Field winding is in series with armature and load. It carries load current and uses few thick turns.

Compound

Both shunt and series fields are used. Cumulative compounding aids flux; differential compounding opposes it.



SF denotes series field. Compound generators may be long-shunt or short-shunt depending on whether the shunt field is connected across both armature and series field or across the armature alone.

Type	Field connection	Voltage behaviour	Typical use
Separately excited	Independent supply	Wide controllability	Laboratory and controlled DC supply
Shunt	Across terminals	Approximately constant voltage	Battery charging, lighting, exciters
Series	Series with load	Voltage rises with load before saturation	Booster and special constant-current work
Cumulative compound	Series field aids shunt field	Flat or rising external characteristic	Distribution feeders, heavy load variation
Differential compound	Series field opposes shunt field	Voltage falls rapidly with load	Rare; special drooping-voltage duty

5. Generator voltage equations

Basic generator equation

$$V = E_g - I_a R_a - V_b$$

If brush drop is neglected, V = Eg − IaRa. State the assumption in numerical work.

Current relations

Shunt: Ia = IL + Ish, Ish = V/Rsh.

Series: Ia = IL; V = Eg − Ia(Ra+Rse).

Long-shunt compound: Ish = V/Rsh; Ia = IL + Ish.

6. Derivation of the DC-generator EMF equation

1. Flux cut by one conductor in one complete revolution = PΦ webers, because it passes under P

poles.

2. Time for one revolution at N rpm = 60/N seconds.

3. Average EMF induced in one conductor = flux cut / time = PΦN/60 volts.

4. Conductors in series in each parallel path = Z/A.

5. Generated EMF between brushes = EMF per conductor × conductors in series.

$$E_g = P\Phi ZN / 60A$$

Symbol	Meaning	SI unit
Eg	Generated EMF	volt
P	Number of poles	dimensionless
Φ	Flux per pole	weber
Z	Total armature conductors	dimensionless
N	Speed	revolutions per minute
A	Parallel paths	dimensionless

$$\text{Simplex lap winding: } A = P \quad | \quad \text{Simplex wave winding: } A = 2$$

Solved numerical M1-1: Generated EMF

Problem: A 4-pole wave-wound generator has 600 conductors, flux 25 mWb/pole and speed 900 rpm. Find Eg.

Given: P=4, Z=600, Φ=0.025 Wb, N=900 rpm, A=2.

Formula: Eg=PΦZN/(60A).

Substitution: Eg=(4×0.025×600×900)/(60×2)=450 V.

Final answer: Generated EMF = **450 V**. The wave winding gives only two parallel paths and therefore higher voltage.

Common mistake: Do not use A=P for a wave winding.

Solved numerical M1-2: Speed from generated EMF

Problem: A 6-pole simplex lap generator has 720 conductors and 30 mWb/pole. Find speed for $E_g=216$ V.

Given: $P=6$, $A=6$, $Z=720$, $\Phi=0.03$ Wb, $E_g=216$ V.

Formula: $N=60AE_g/(P\Phi Z)$.

Calculation: $N=(60\times6\times216)/(6\times0.03\times720)=600$ rpm.

Final answer: 600 rpm.

7. Losses, power stages and efficiency

Copper losses

Armature: $I_a^2R_a$. Shunt field: V^2/R_{sh} or VI_{sh}. Series field: $I_a^2R_{se}$. Brush-contact loss may be taken as V_bI_a when specified.

Iron losses

Hysteresis and eddy-current losses occur mainly in the armature core because its flux direction changes as it rotates.

Mechanical losses

Bearing friction, brush friction and windage. They are approximately constant over the normal operating range.

Stray-load loss

Additional loss caused by leakage flux, armature reaction and nonuniform current distribution. It is often estimated as a small percentage.

Constant losses	Variable losses
Iron loss, mechanical loss and approximately constant shunt-field loss	Mainly armature and series-field copper loss proportional to load-current squared

Mechanical input
$P_{mech,in}$

↓

Subtract rotational losses
iron + mechanical

↓

Gross electrical power
E_gI_a

↓

Subtract electrical losses
$I_a^2R_a$ + field + brush

↓

Terminal output
VI_L

η_g = Electrical output / Mechanical input × 100%

8. Condition for maximum efficiency

1. Let terminal voltage be approximately constant, load current be I and effective variable resistance be R. Let constant loss be Wc.
2. Output = VI. Total loss = I^2R + Wc.
3. Efficiency η = VI / (VI + I^2R + Wc).
4. For maximum efficiency, differentiate η with respect to I and set dη/dI=0.
5. After simplification: I^2R = Wc.

At maximum efficiency: variable copper loss = constant loss

This result assumes approximately constant terminal voltage, constant field loss and constant rotational losses. For a shunt generator, armature current rather than load current should be used when field current is appreciable.

Solved numerical M1-3: Load at maximum efficiency

Problem: A 220 V shunt generator has armature resistance 0.20 Ω and constant losses 800 W. Neglect field-current difference. Find load current at maximum efficiency.

Formula: $I^2Ra=Wc$.

Calculation: $I=\sqrt{(800/0.20)}=\sqrt{4000}=63.25$ A.

Output: 220×63.25=13.915 kW.

Final answer: Maximum efficiency occurs at approximately **63.3 A** load current.

9. Armature winding fundamentals

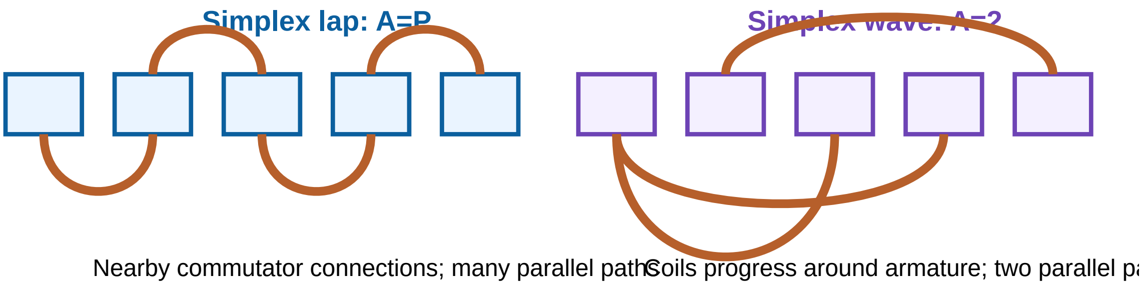
A coil has one or more turns. Each turn has two active coil sides placed in slots. Coil ends connect to commutator segments. The winding must provide closed paths and correctly spaced connections so generated EMFs add.

Simplex lap winding

Each coil “laps back” to a nearby commutator segment. Parallel paths A=P. Brush sets are normally equal to pole number. Suitable for low-voltage, high-current machines.

Simplex wave winding

The winding progresses around the armature in a wave, connecting conductors under poles of similar polarity. Parallel paths A=2. Suitable for high-voltage, lower-current machines.



Conceptual path only: practical winding diagrams use coil-span, back pitch, front pitch and commutator pitch.

Feature	Lap winding	Wave winding
Parallel paths, simplex	A=P	A=2
Brush sets	Usually P	Usually 2
Voltage/current suitability	Low voltage, high current	High voltage, low current
Equalizer rings	Required to prevent circulating current	Normally not required
Dummy coil	Not normally required	May be required to satisfy pitch conditions
Typical use	Heavy-current generators	Lighting and higher-voltage generators

Module 1 expected-question practice

One word

Name the part acting as a mechanical rectifier.

Answer: Commutator.

Important diagram

Draw and label DC-machine construction.

Include yoke, pole core, pole shoe, field winding, armature, commutator, brushes, shaft and bearings.

Important derivation

Derive $E_g = P\Phi ZN/60A$.

Use flux cut per revolution, time per revolution and conductors in series per path.

Important numerical

Find E_g, N, Φ or Z from the EMF equation.

First identify lap/wave winding and correct A.

Comparison

Compare lap and wave windings.

Parallel paths, brushes, current/voltage rating and equalizer rings.

Long answer

Explain losses, power flow and maximum efficiency.

Classify losses, draw stages and derive $I^2R = W_c$.

MODULE 2 • CO2 • 10 HOURS + SERIES TEST I

Characteristics and Applications of DC Generators

M2.01–M2.04: armature reaction, commutation, generator characteristics, voltage build-up, parallel operation and applications.

1. Armature reaction

When load current flows in armature conductors, the armature produces its own magnetic field. This armature field combines with the main pole field and distorts the air-gap flux distribution. The effect of armature magnetomotive force on the main field is called **armature reaction**.

Geometrical neutral axis (GNA)

The axis midway between adjacent pole centres. It is fixed by machine geometry.

Magnetic neutral axis (MNA)

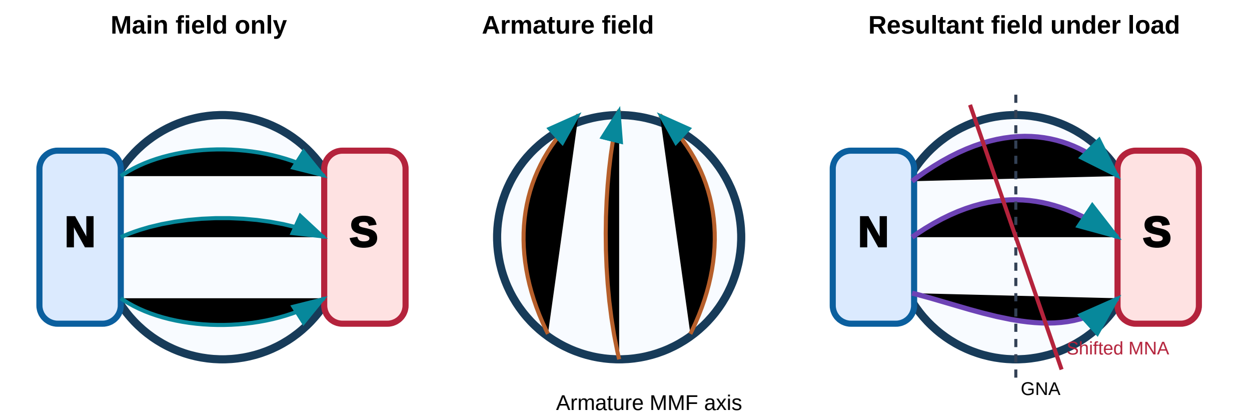
The axis along which armature conductors have zero induced EMF. Under load it shifts because the resultant field is distorted.

Cross-magnetising effect

Armature MMF acts mainly across the pole axis, strengthening flux at one pole tip and weakening it at the other.

Demagnetising effect

If brushes are shifted from the GNA, part of armature MMF directly opposes the main field and reduces average flux.



Flux crowds at one pole tip and weakens at the other. In a generator, MNA shifts in the direction of rotation. Brush shifting alone is unsatisfactory because the required shift varies with load.

Effects of armature reaction

Distorted flux, reduced average flux, lower generated EMF, shifted MNA, brush sparking, local heating at pole tips and poor commutation.

Malayalam Support: □□□□ □□□□□□□ □□□□ armature □□□□□□□□ magnetic field □□□□□□□□□□□□□□. □□□ main field-□□ □□□□□□□□□□□□□□□□□□□ neutral axis □□□□□□□□□□□ □□□□□□□□□□; □□□□□□□ brush sparking □□□□□□□□□.

2. Interpoles and compensating windings

Interpoles

Small poles placed midway between main poles. Their windings are connected in series with the armature, so interpole flux automatically varies with load. In a generator, interpole polarity is the same as the next main pole in the direction of rotation. Interpoles induce a commutating EMF that opposes reactance voltage.

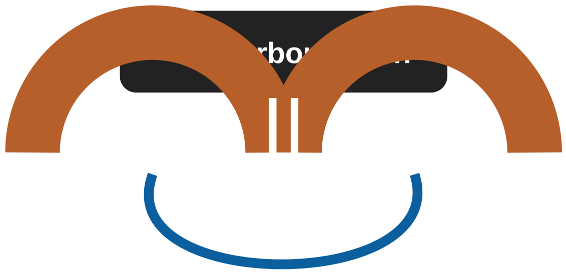
Compensating winding

Conductors embedded in slots on the pole faces and connected in series with the armature. Their MMF opposes armature MMF under the pole arc, keeping air-gap flux nearly uniform during rapid load changes.

Feature	Interpoles	Compensating winding
Location	Between main poles	In main-pole shoes
Main purpose	Improve commutation in neutral zone	Neutralise cross-magnetising effect under pole arc
Connection	Series with armature	Series with armature
Use	Common in medium/large machines	Heavy-duty machines with rapidly changing load

3. Commutation

Commutation is the reversal of current in an armature coil while the brush short-circuits the two commutator segments connected to that coil. Ideally, current changes linearly from +I to −I during the commutation period and reaches its final value exactly when the coil leaves the brush.



Coil is short-circuited while brush bridges two segments.

Current must reverse from +I to −I

Inductance opposes current reversal and produces reactance voltage $e_r = L \cdot di/dt$. If reversal is incomplete, current interruption at the brush edge causes sparking.

Ideal commutation

Current reverses uniformly and reaches −I by the end of brush contact. No residual current is interrupted.

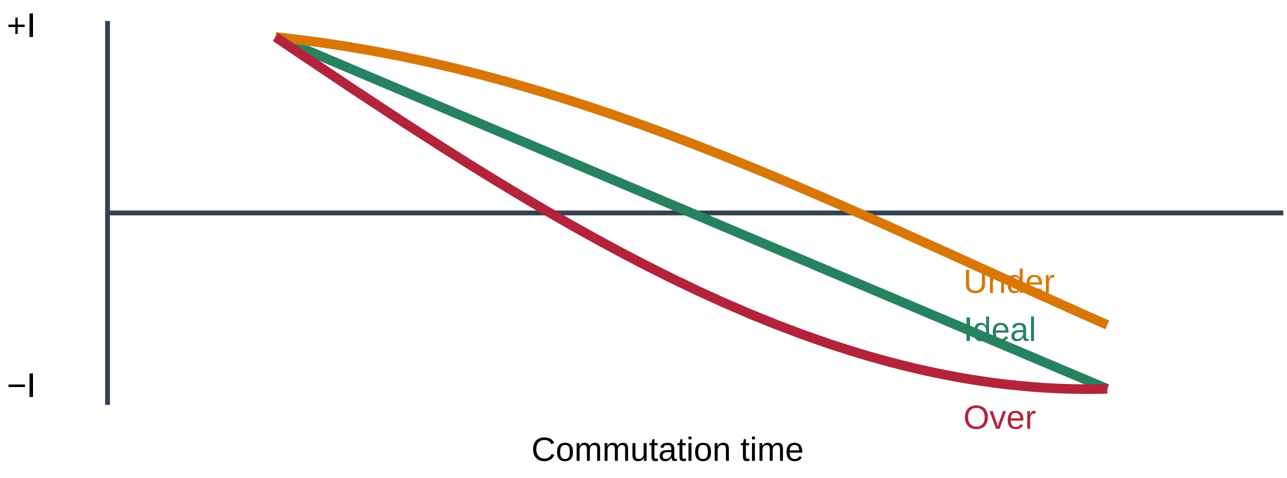
Under-commutation

Current reverses too slowly. At brush exit, current has not reached −I, causing sparking at the trailing edge.

Over-commutation

Current reverses too quickly and overshoots. Sparking may occur at the leading edge.

Current reversal during commutation



Methods of improving commutation

Carbon brushes

Higher contact resistance helps current transfer gradually from one segment to the next: resistance commutation.

Interpoles

Provide reversing EMF proportional to armature current: EMF commutation.

Correct brush position

Brushes should lie on the actual MNA, though fixed brush shift is inadequate over wide load range.

Shorter coil inductance

Good winding design reduces reactance voltage and improves current reversal.

4. Equalizer connections

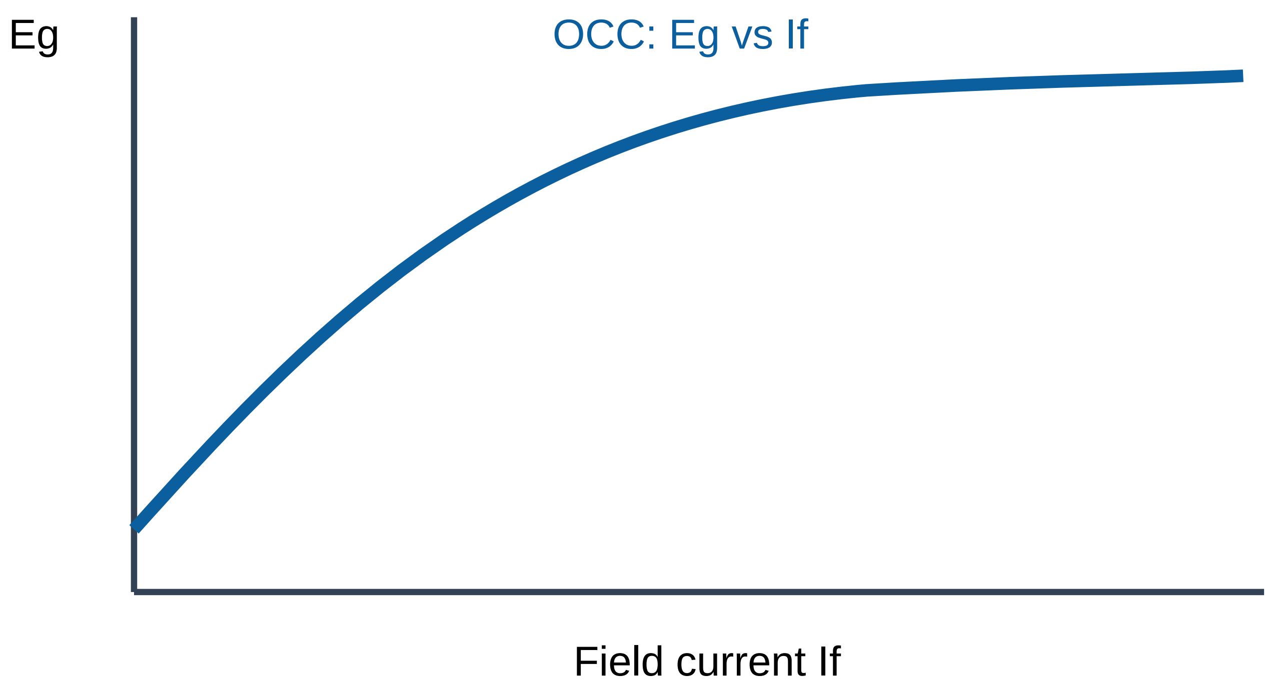
In a lap winding, each pole provides a separate parallel path. Small differences in air-gap flux can produce unequal path EMFs, causing circulating currents through brushes and commutator. Equalizer rings connect equipotential points of the winding, allowing circulating current to flow inside the armature instead of through the brushes. Wave windings normally do not require equalizer rings because each parallel path includes conductors under all poles.

5. Generator characteristics

A characteristic is a graph showing how voltage, current or excitation changes under specified operating conditions.

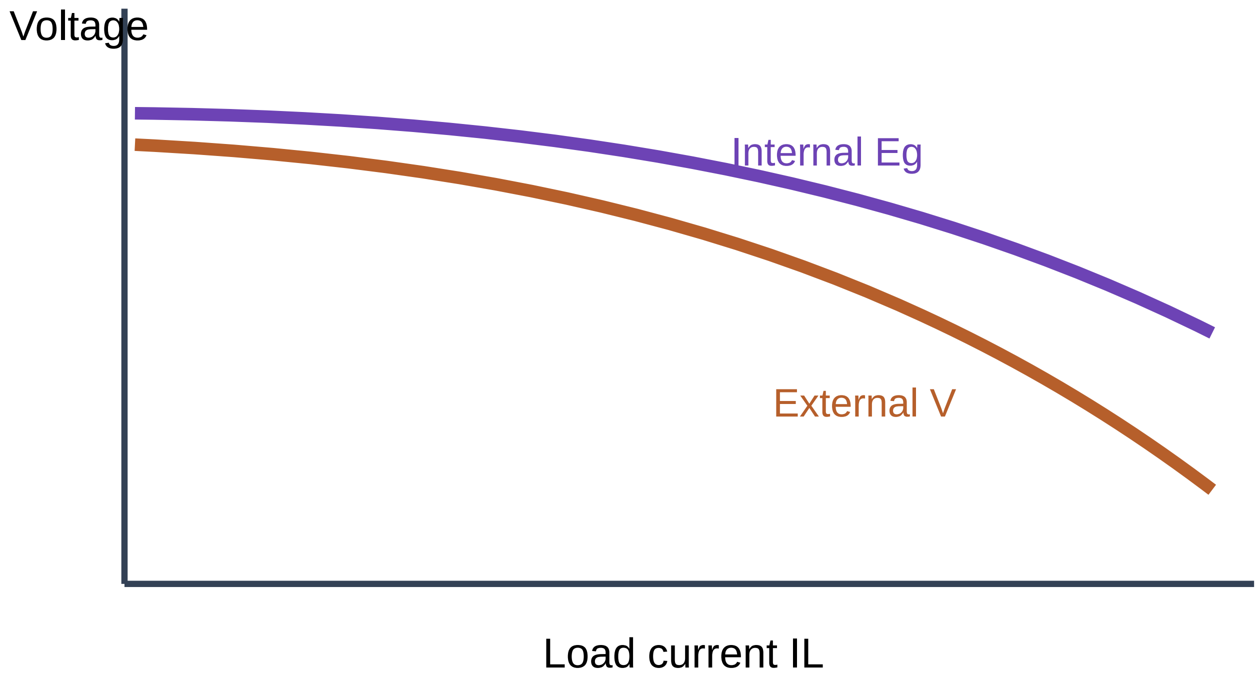
Characteristic	Axes	Meaning
Open-circuit characteristic (OCC)	Eg versus field current If at constant speed, no load	Magnetisation curve; shows residual voltage and saturation
Internal characteristic	Generated EMF after armature-reaction effect versus load current	Lies below ideal generated-voltage relation
External characteristic	Terminal voltage V versus load current IL	Includes armature-reaction and internal voltage drops

Separately excited generator



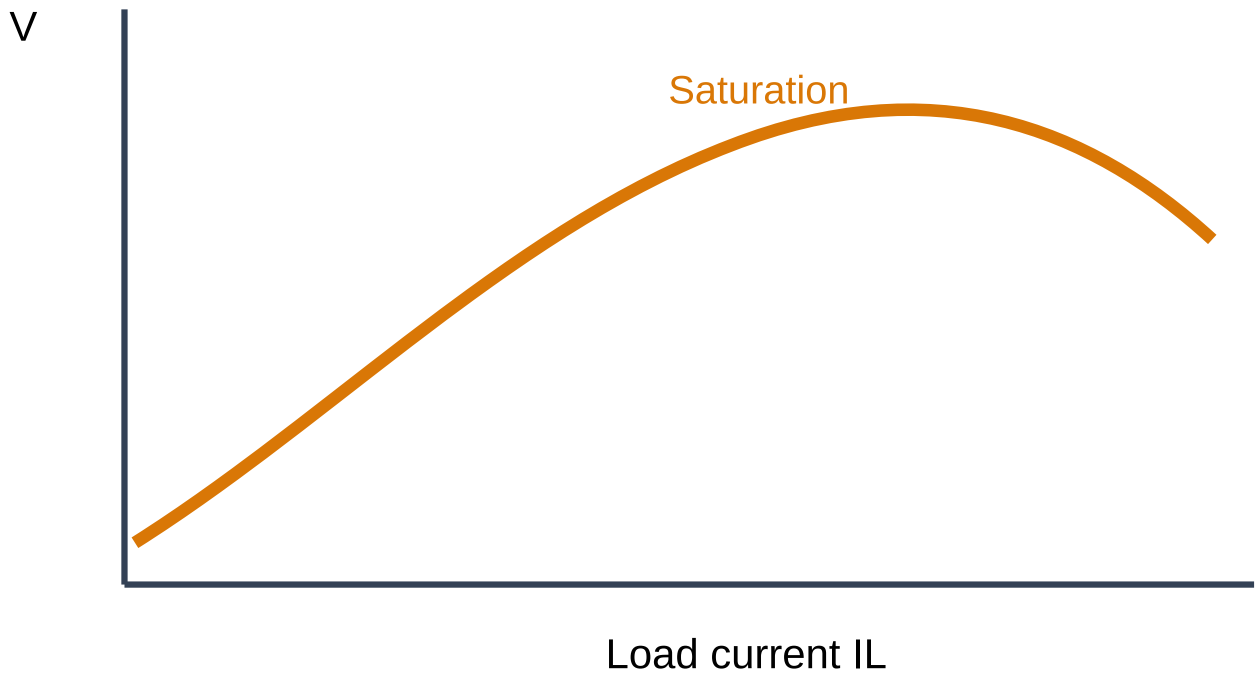
At first, Eg rises almost linearly with If. Magnetic saturation bends the curve.

Shunt generator: internal and external



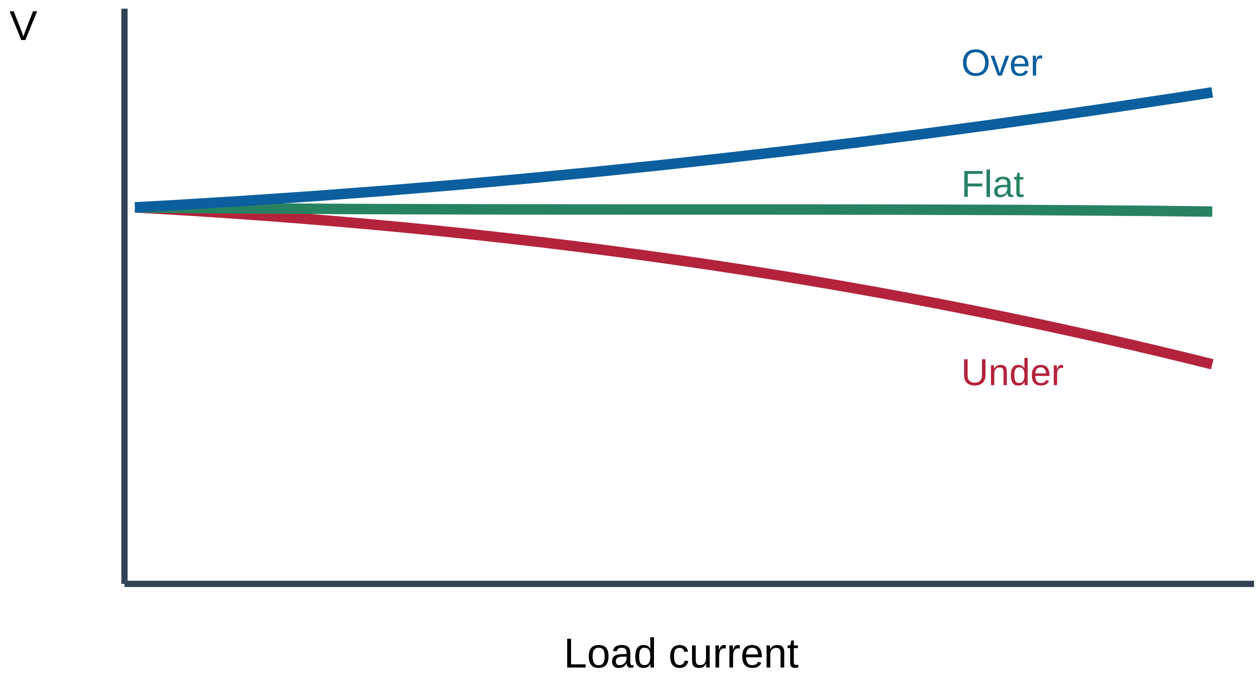
Terminal voltage falls because IaRa drop and armature reaction reduce voltage. Falling V also reduces shunt-field current, causing a cumulative drop.

Series generator



Voltage initially rises with load as series-field flux increases, then saturates and may fall because of internal drops and armature reaction.

Compound generators



6. Voltage build-up of a shunt generator

1. Residual magnetism produces a small initial EMF when the armature starts rotating.
2. This EMF sends current through the shunt field.
3. Field current strengthens flux in the same direction as residual flux.
4. Higher flux produces higher Eg, which produces more field current.
5. Voltage rises cumulatively until the field-resistance line intersects the OCC.

Malayalam Support: Residual magnetism □□□□ □□□□□ voltage □□□□□ □□□□□□□□. □□□□□□□ field current □□□□□ flux □□□□□□□; flux □□□□□□□□□ generated voltage □□□□□□□ □□□□□. □□□□□□ voltage build up □□□□□□□□□□.

Conditions for build-up

- Residual magnetism must exist.
- Field connection must aid residual flux.
- Field-circuit resistance must be below critical resistance.
- Speed must exceed critical speed for the chosen field resistance.
- Field circuit must be closed.

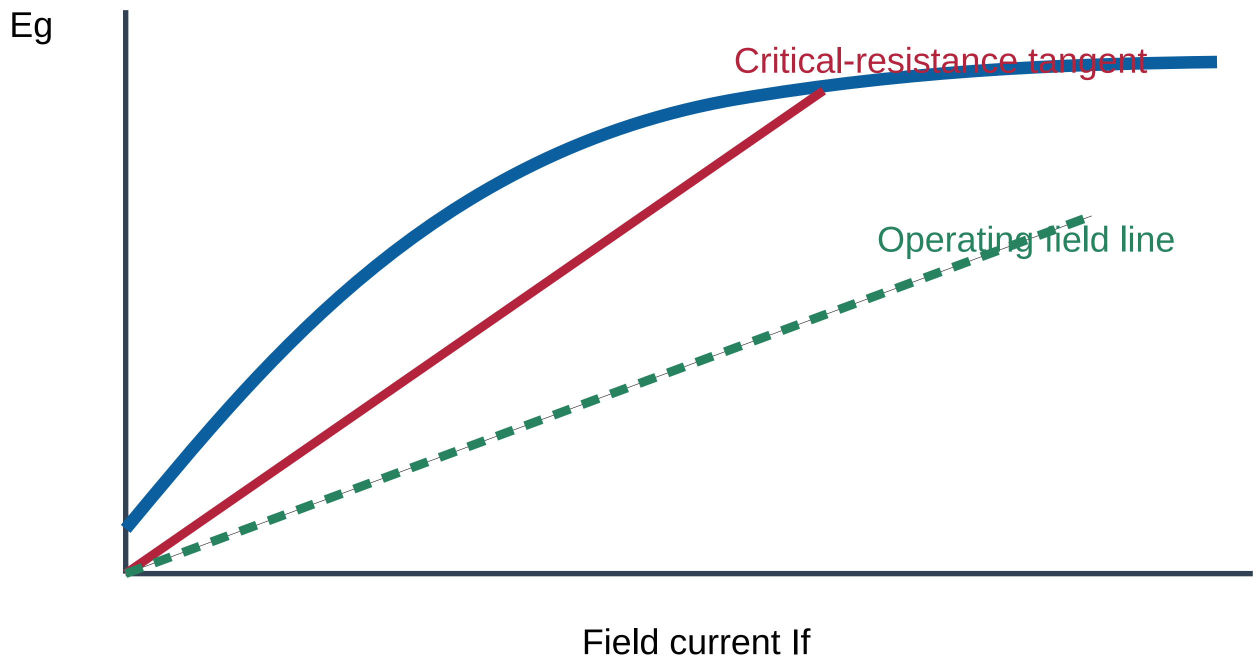
Failure causes

- Loss of residual magnetism.
- Reversed field connection.
- Excessive field resistance.
- Speed too low.
- Open field circuit or poor brush contact.

7. Critical resistance and critical speed

Critical field resistance is the maximum shunt-field resistance at which a generator just builds voltage at a given speed. Graphically it is the slope of the line drawn from the origin tangent to the OCC. **Critical speed** is the minimum speed at which the machine builds voltage for a specified field resistance.

Critical-resistance tangent method



Solved numerical M2-1: Critical resistance from OCC data

Problem: The tangent to a shunt-generator OCC passes through If=2.0 A, Eg=220 V. Determine critical field resistance.

Formula: Rcrit = slope = Eg/If.

Calculation: Rcrit=220/2=110 Ω.

Final answer: 110 Ω. The actual field-circuit resistance must be less than this at that speed.

Solved numerical M2-2: Critical speed

Problem: At 1000 rpm the critical resistance is 120 Ω. Assuming the unsaturated OCC is proportional to speed, estimate critical speed for a 90 Ω field circuit.

Relation: Critical-resistance slope is proportional to speed, so $N_{crit}/N_1 = R_f/R_{crit,1}$.

Calculation: $N_{crit} = 1000 \times 90/120 = 750$ rpm.

Final answer: 750 rpm.

Assumption: Use the linear/unsaturated portion and geometrically similar OCC.

8. Parallel operation of DC generators

Parallel operation improves reliability, permits maintenance without total shutdown, allows economical matching of generating capacity to load and provides reserve capacity. Before closing the parallel switch, polarity must be the same and incoming voltage must approximately equal busbar voltage.

1. Run the incoming generator at rated speed with its switch open.
2. Verify correct polarity.
3. Adjust excitation until incoming voltage equals bus voltage.
4. Close the switch.
5. Increase incoming-generator excitation to make it take load; adjust machines for desired sharing.

Safety note

Wrong polarity effectively short-circuits the generators through the busbars. Never parallel without polarity and voltage checks.

Equalizer connections are essential for compound generators so unequal series-field currents do not cause one generator to take excessive load.

9. Application selection

Requirement	Suitable generator	Reason
Adjustable DC laboratory supply	Separately excited	Field can be controlled independently
Nearly constant-voltage supply	Shunt generator	Small voltage drop with load
Feeder-voltage compensation / booster	Series generator	Voltage increases with load current
Constant terminal voltage with changing load	Flat-compounded generator	Series field offsets internal voltage drop
Rising voltage at remote load	Over-compounded generator	Compensates both machine and feeder drop

Module 2 expected-question practice

Definition

Define armature reaction and MNA.

State armature-field effect and zero-EMF axis.

Important diagram

Show main, armature and resultant flux.

Label GNA, shifted MNA and crowded pole tip.

Comparison

Interpoles versus compensating windings.

Compare location, purpose and connection.

Characteristic curve

Draw OCC, internal and external curves.

Label axes and explain why each curve lies below the previous one.

Important numerical

Find critical resistance or critical speed.

Use tangent slope and proportionality with speed.

Long answer

Explain voltage build-up and failure causes.

Include residual magnetism, polarity, resistance and speed conditions.

MODULE 3 • CO3 • 12 HOURS

DC Motors, Starters and Testing

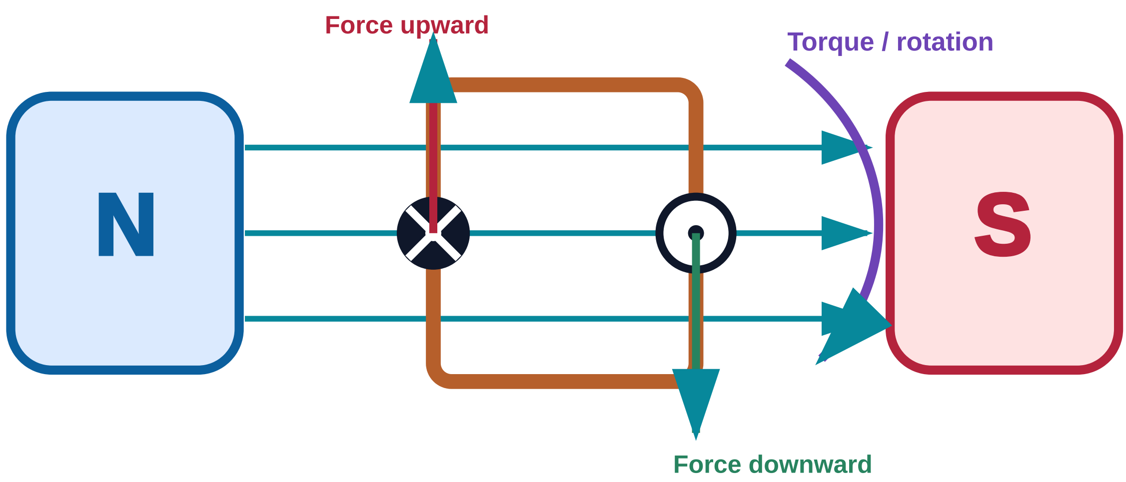
M3.01–M3.04: motor principle, back EMF, torque, performance characteristics, starters, load test and Swinburne's test.

1. Working principle of a DC motor

A current-carrying conductor placed in a magnetic field experiences a force. In a DC motor, forces on armature conductors form a couple and produce torque. Fleming's left-hand rule gives the force direction: first finger points along field, second finger along current and thumb along force or motion.

$$F = B\,I\,l\,\sin\theta$$

F in newtons; B in tesla; I in amperes; active length l in metres.



Cross and dot symbols show current into and out of the page. Opposite forces form a turning couple. The commutator reverses conductor current when a coil moves under the next pole so torque remains in the same direction.

Malayalam Support: □□□□□□□□ □□□□□□□ current □□□□□□□□□ conductor-□□ force □□□□□□□□□□□□. Armature-□□□□ □□□□□□ □□□□□□□□□□ □□□□□□□□ force-□□ □□□□□□□□ torque □□□□□□□□□□□□□□.

2. Back EMF and self-regulation

As the armature rotates, its conductors cut flux and generate an EMF by generator action. By Lenz's law, this induced EMF opposes the applied voltage, so it is called **back EMF** Eb. It limits armature current during normal running.

$$V = E_b + I_aR_a + V_b$$

Neglecting brush drop: $E_b = V - I_aR_a$

$$E_b = P\Phi ZN / 60A$$

Self-regulating action

If load increases, speed falls slightly. Eb falls, Ia=(V−Eb)/Ra rises, torque rises and the motor meets the larger load. If load decreases, speed and Eb rise, reducing Ia and torque.

Malayalam Support: Speed □□□□□□□□□□ back EMF □□□□□; □□□□□□□ armature current □□□□□□□. Load □□□□□□□□ speed □□□□□ □□□□□□□□□□ back EMF □□□□□□□□□□ current/torque □□□□□□□□□ □□□□□□□□.

3. Why a starter is essential

At starting, N=0 and therefore Eb=0. Because armature resistance is very small, connecting the motor directly to supply would cause a dangerously high current.

$$I_{a,start} = V / R_a \quad \text{(without starter)}$$

$$I_{a,start} = V / (R_a + R_{st}) \quad \text{(with starter resistance)}$$

Solved numerical M3-1: Starting current and starter resistance

Problem: A 220 V motor has Ra=0.25 Ω. Find direct starting current and series resistance needed to limit current to 50 A.

Direct start: Ia=220/0.25=880 A.

Required total resistance: Rtotal=220/50=4.4 Ω.

Starter resistance: Rst=4.4−0.25=4.15 Ω.

Final answer: Direct current **880 A**; starter resistance **4.15 Ω**.

4. Classification and connection

Separately excited

Field supplied independently. Excellent independent flux control.

Shunt motor

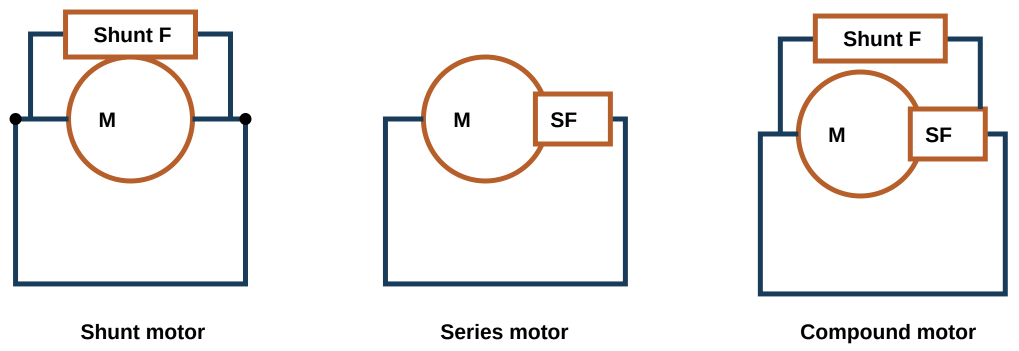
Field connected across supply. Nearly constant flux and speed.

Series motor

Field in series with armature. Flux initially proportional to Ia; very high starting torque.

Compound motor

Shunt and series fields combine. Cumulative compound gives high starting torque with better speed regulation.



In a cumulative compound motor, series-field MMF aids the shunt field. Differential compounding is generally unstable for motor duty and is rarely used.

5. Mechanical power and torque

Armature electrical input is VIa. Armature copper loss is Ia²Ra. The remaining power converted electromagnetically is called gross mechanical power developed.

$$P_m = E_b I_a$$

$$\omega = 2\pi N/60 \text{ rad/s}$$

$$T_a = P_m/\omega$$

Derivation of the general torque equation

1. Mechanical power developed Pm = EbIa.
2. Substitute Eb=PΦZN/(60A).
3. Angular speed ω=2πN/60.
4. Torque Ta=Pm/ω=[PΦZN Ia/(60A)]/[2πN/60].
5. Cancel N and simplify.

$$T_a = (PZ/2\pi A)\,\Phi I_a = 0.159(PZ/A)\Phi I_a \text{ N·m}$$

$$\text{Therefore } T \propto \Phi I_a$$

Armature torque is the electromagnetic torque developed inside the armature. **Shaft torque** is the useful output torque after subtracting torque equivalent of iron and mechanical losses.

$$T_{sh} = P_{out}/\omega$$

Solved numerical M3-2: Back EMF, developed power and torque

Problem: A 220 V shunt motor takes armature current 40 A. Ra=0.30 Ω and speed=1000 rpm. Neglect brush drop. Find Eb, Pm and Ta.

Back EMF: Eb=220−40×0.30=208 V.

Developed power: Pm=208×40=8320 W.

Angular speed: ω=2π×1000/60=104.72 rad/s.

Torque: Ta=8320/104.72=79.45 N·m.

Final answer: Eb=**208 V**, Pm=**8.32 kW**, Ta=**79.5 N·m**.

6. Speed relation

$$N \propto E_b/\Phi = (V - I_a R_a)/\Phi$$

Speed can therefore be changed by varying flux, armature-circuit resistance or applied armature voltage. This relation is the foundation of Module 4.

7. Condition for maximum mechanical power

1. Ignoring brush drop, Ia=(V−Eb)/Ra.
2. Mechanical power developed Pm=EbIa=Eb(V−Eb)/Ra.
3. Differentiate with respect to Eb: dPm/dEb=(V−2Eb)/Ra.
4. Set derivative to zero: V−2Eb=0.

$$\text{At maximum mechanical power: } E_b = V/2$$

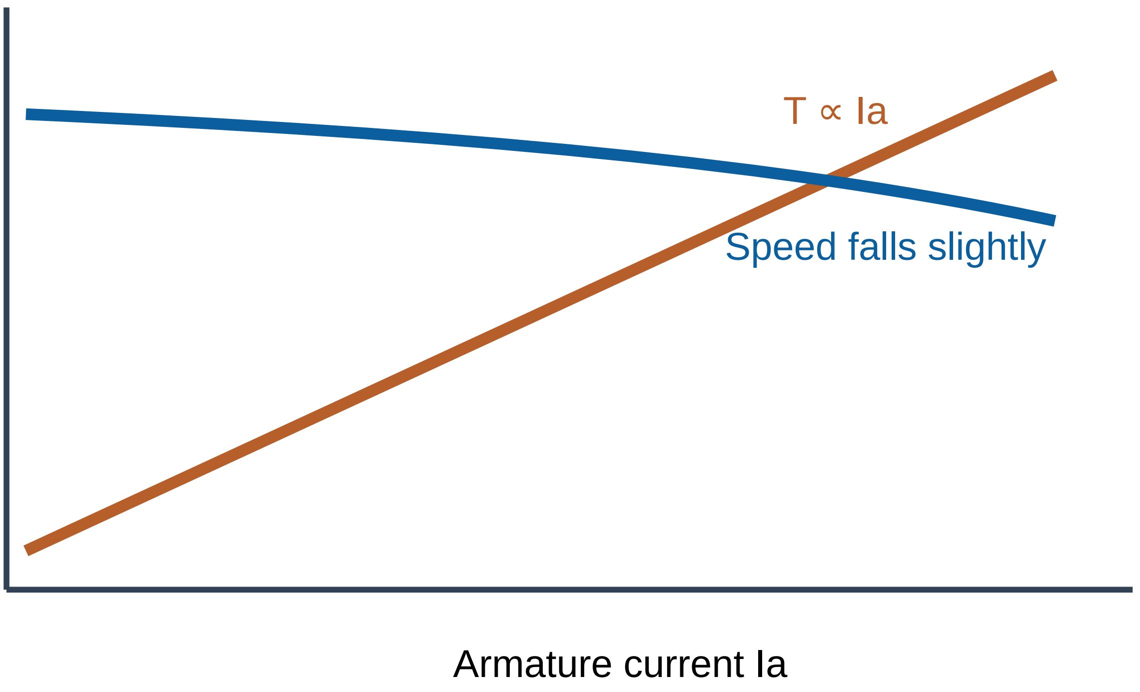
Why this is not normal operation

At Eb=V/2, Ia=V/(2Ra). Armature copper loss Ia^2Ra equals mechanical power developed, so even before rotational losses the efficiency is only 50%. Current is excessive and heating is severe.

8. Motor characteristics

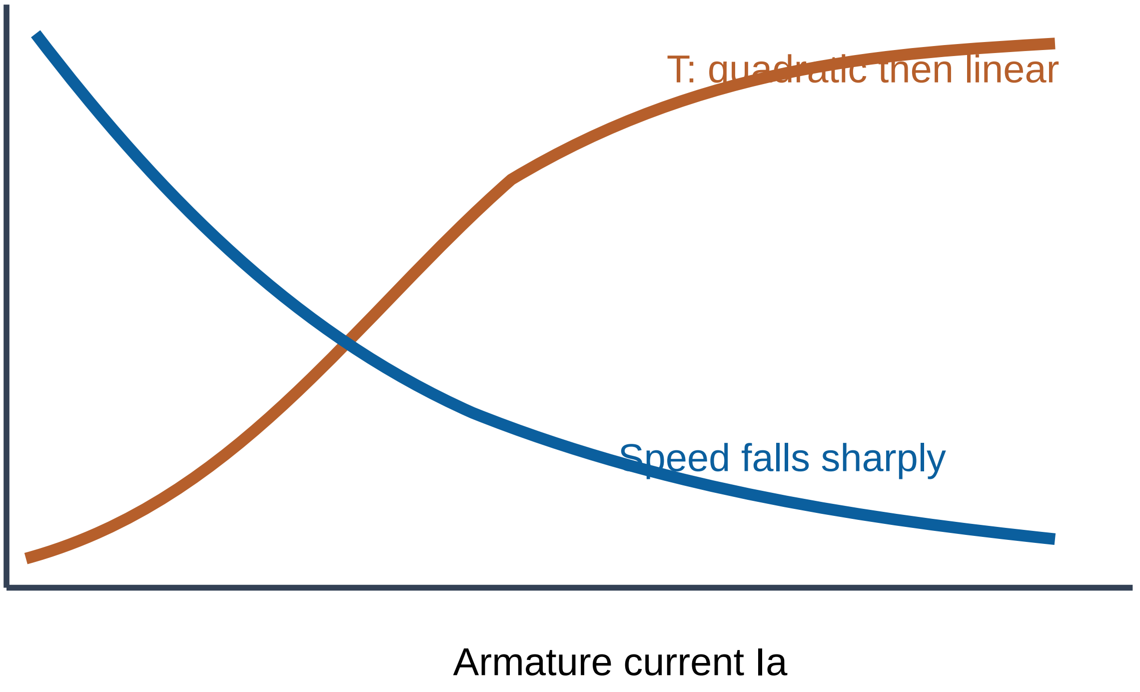
Electrical characteristic usually means torque versus armature current. Mechanical characteristic means speed versus torque. Performance characteristic includes speed versus current, efficiency versus output and other operating curves.

Shunt motor characteristics



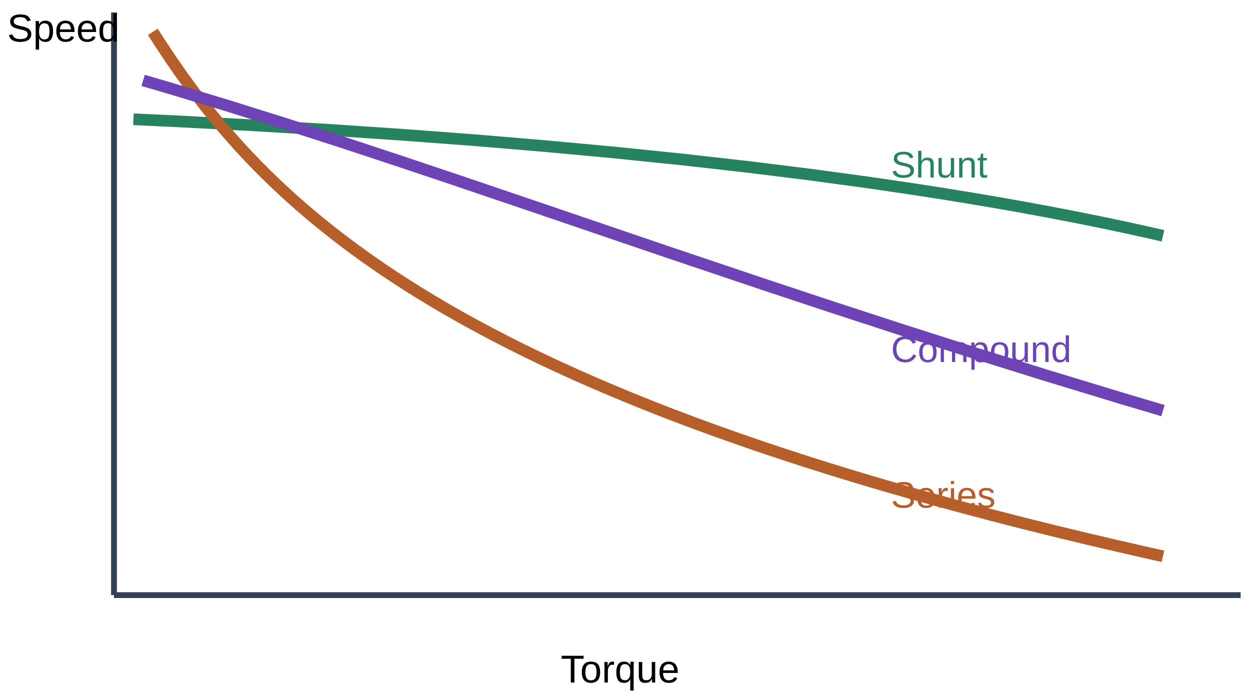
Flux is approximately constant: torque is proportional to Ia and speed is nearly constant.

Series motor characteristics



Before saturation, $\Phi \propto Ia$ so $T \propto Ia^2$. At light load, flux is small and speed can become dangerously high; never run a series motor without load.

Speed–torque comparison



Application selection logic

- **Nearly constant speed:** shunt motor.
- **Very high starting torque:** series motor.
- **High starting torque plus better speed regulation:** cumulative compound motor.
- **Independent wide control:** separately excited motor.

Motor	Starting torque	Speed regulation	Suitable applications
Shunt	Moderate	Good	Lathes, fans, blowers, pumps, machine tools
Series	Very high	Poor; no-load dangerous	Cranes, hoists, traction, elevators
Cumulative compound	High	Better than series	Conveyors, presses, rolling mills, reciprocating pumps
Separately excited	Controlled	Excellent with feedback/control	Precision drives, laboratories, variable-speed systems

9. Motor losses, power flow and efficiency

Electrical input
VIL

↓

Subtract field & armature copper losses
I^2R + brush

↓

Gross mechanical power
EbIa

↓

Subtract rotational & stray losses	↓	Shaft output Pout

η_m = Mechanical output / Electrical input × 100%

At maximum efficiency: variable copper loss = constant loss

Use the correct current and include field loss consistently.

Solved numerical M3-3: Motor efficiency

Problem: A 220 V shunt motor takes line current 30 A. Shunt-field current is 1 A, $R_a=0.4\ \Omega$ and rotational losses are 500 W. Find shaft output and efficiency.

Armature current: $I_a=30-1=29$ A.

Input: $P_{in}=220\times30=6600$ W.

Armature copper loss: $29^2\times0.4=336.4$ W.

Field loss: $220\times1=220$ W.

Shaft output: $6600-336.4-220-500=5543.6$ W.

Efficiency: $5543.6/6600\times100=83.99\%$.

Final answer: Output ≈**5.54 kW**; efficiency ≈**84.0%**.

10. Three-point starter

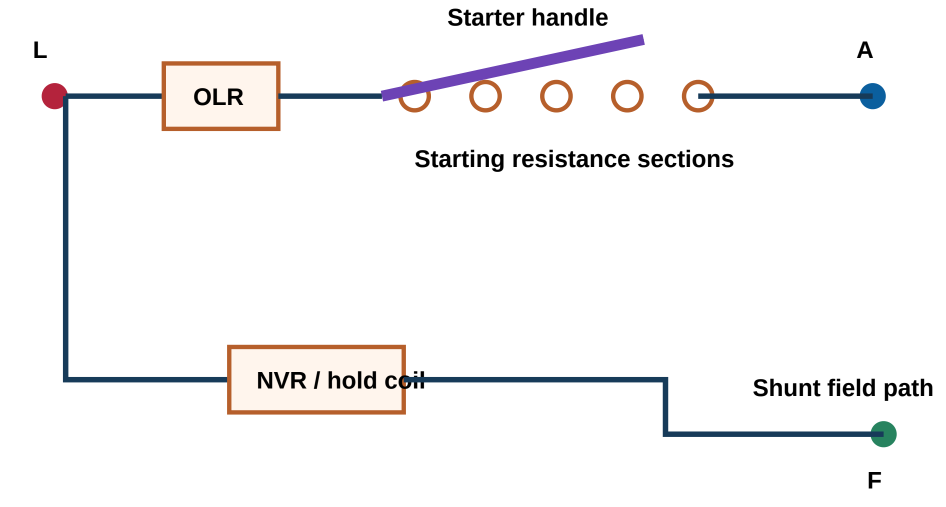
The three terminals are L (line), A (armature) and F (field). Starting resistance is divided into sections. Moving the handle from OFF toward RUN progressively removes resistance as speed and back EMF rise.

No-volt release (NVR)

Connected in series with the shunt field. It magnetically holds the handle in RUN. Supply failure or field-circuit opening releases the handle and spring returns it to OFF.

Overload release (OLR)

Connected in series with armature. Excess current energises it and releases the NVR mechanism, disconnecting the motor.



Because the NVR is in the field circuit, weakening the shunt field for speed control may reduce holding current and unintentionally release the handle.

11. Four-point starter

The four terminals are L, A, F and N. The NVR coil is connected directly across the supply through a limiting resistance, independent of the shunt-field circuit. Therefore field weakening does not release the handle. However, this arrangement does not directly protect against field-circuit opening in the same way as the three-point starter.

Feature	Three-point starter	Four-point starter
Terminals	L, A, F	L, A, F, N
NVR connection	Series with shunt field	Across supply independently
Field-control compatibility	May release at weak field	Suitable for field weakening
Field-failure protection	Better inherent protection	Needs separate field-failure protection

12. Direct load test

In a direct load test, the motor drives a brake, generator or dynamometer. Measure supply voltage, line/armature current, speed and output torque. Then calculate input, output and efficiency at different loads.

$$P_{out} = 2\pi NT/60$$

1. Start the motor with no load using the starter.
2. Record no-load readings.
3. Increase load gradually and record V, I, N and torque.
4. Compute input, output and efficiency.
5. Do not exceed rated current or temperature.

Laboratory safety
Check couplings, guards, earthing and instrument ranges. Never touch rotating parts or adjust a mechanical brake without supervision.

13. Swinburne's test

Swinburne's test is an indirect no-load test mainly for DC shunt and compound machines. Run the motor at rated voltage and speed without mechanical load. The no-load input supplies armature copper loss, field loss and constant rotational losses. From these, efficiency at any assumed load can be predicted.

1. Measure supply voltage V, no-load line current $I0$, field current Ish and armature resistance Ra.
2. No-load armature current $Ia0=I0-Ish$.
3. No-load input $P0=VI0$.
4. Constant rotational loss $Wc=P0-VIsh-Ia0^2Ra$ (subtract brush loss if specified).
5. At a chosen load, calculate armature copper loss and combine with Wc and field loss to predict efficiency.

Advantages

- Simple and economical.
- Requires only no-load power.
- Efficiency can be predicted for motor and generator operation.
- No large loading arrangement required.

Limitations

- Temperature rise under full load is not tested.
- Stray-load loss and armature-reaction effects are not accurately included.
- Not suitable for series motors.
- Commutation under load is not assessed.

Solved numerical M3-4: Swinburne's-test prediction

Problem: A 220 V shunt motor takes 5 A at no load. Field current=1 A and Ra=0.5 Ω. Predict efficiency when line current is 40 A.

No-load armature current: $I_{a0}=5-1=4$ A.

No-load input: $220\times5=1100$ W.

Field loss: $220\times1=220$ W.

No-load armature copper: $4^2\times0.5=8$ W.

Constant rotational loss: $1100-220-8=872$ W.

At 40 A line current: $I_a=39$ A; armature copper$=39^2\times0.5=760.5$ W.

Total losses: $872+220+760.5=1852.5$ W.

Input: $220\times40=8800$ W; output$=6947.5$ W.

Efficiency: $6947.5/8800\times100=78.95\%$.

Final answer: Predicted motor efficiency ≈**79.0%**.

Feature	Direct load test	Swinburne's test
Method	Actual mechanical loading	No-load indirect prediction
Power required	High at full load	Low
Temperature and commutation	Observed under load	Not accurately assessed
Efficiency accuracy	Actual at tested load	Predicted
Suitable machines	Most motors if loading equipment available	Shunt and cumulative compound; not series

Module 3 expected-question practice

Very short

Why is a starter required?

At start $E_b=0$; small R_a would cause excessive current.

Important derivation

Derive $T=0.159(PZ/A)\Phi I_a$.

Use $P_m=E_bI_a$ and $T_a=P_m/\omega$.

Important derivation

Derive maximum-power condition.

Differentiate $E_b(V-E_b)/R_a$ to obtain $E_b=V/2$ and explain low efficiency.

Characteristic curve

Draw shunt and series motor curves.

Show torque-current, speed-current and speed-torque relations.

Important diagram

Explain three-point and four-point starters.

Label starting resistance, handle, NVR and OLR.

Long answer

Explain Swinburne's test.

Procedure, calculation, advantages, limitations and prediction numerical.

MODULE 4 • CO4 • 11 HOURS + SERIES TEST II

Speed Control and Traction Motors

M4.01–M4.04: speed-control methods, DC series motors in traction, series-parallel control, shunt/bridge transitions and electrical braking.

1. Speed equation and control variables

$$N \propto E_b/\Phi = (V - I_a R_a)/\Phi$$

Motor speed is controlled by changing applied armature voltage V, armature-circuit resistance, or flux per pole Φ. The practical method depends on motor type, required speed range, efficiency, load torque and cost.

Flux control

Reduce flux to obtain speed above base speed. Efficient because field current is small, but excessive weakening reduces torque capability and may cause overspeed.

Armature resistance control

Insert series resistance to reduce Eb and speed below base speed. Simple but inefficient; speed varies with load.

Voltage control

Vary armature voltage while maintaining flux. Gives smooth, efficient control below base speed using controlled rectifiers, choppers or motor-generator systems.

Method	Speed range	Efficiency	Load regulation	Main limitation
Flux control	Above base speed	High	Good if field is stable	Reduced torque and commutation margin
Armature resistance	Below base speed	Low	Poor	Large I^2R loss and load-dependent speed
Voltage control	Mostly below base speed	High	Good	Needs variable DC source/power electronics

2. Shunt-motor speed control

Field-rheostat control

A rheostat in series with the shunt field reduces field current and flux. Since $N \propto 1/\Phi$ approximately, speed rises. It is used above rated/base speed.

For approximately constant Eb: $N_1/N_2 = \Phi_2/\Phi_1$

Armature-resistance control

A resistor in series with armature increases voltage drop IaR, reducing Eb and speed. Used for short-duration low-speed duty.

$N_2/N_1 = [(V-Ia(Ra+Rext))/\Phi_2] / [(V-IaRa)/\Phi_1]$

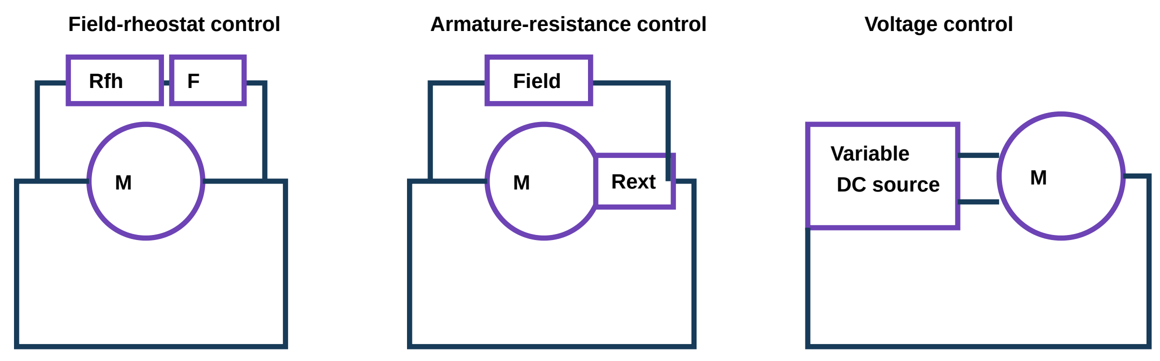
Armature-voltage control

With constant flux, speed is nearly proportional to armature voltage. Modern chopper drives provide smooth control and regenerative capability.

For constant Φ: $N_1/N_2 = E_{b1}/E_{b2}$

Constant-torque / constant-power regions

Below base speed, voltage control with rated flux gives nearly constant torque. Above base speed, field weakening reduces flux and available torque, so output is approximately constant power within limits.



Field control changes Φ; armature resistance wastes power in Rext; voltage control changes armature terminal voltage with better efficiency.

Solved numerical M4-1: Speed after flux change

Problem: A shunt motor runs at 1000 rpm with flux $\Phi 1$. If field flux is reduced by 20% and armature voltage drop is unchanged, find new speed.

Given: $\Phi 2=0.8\Phi 1$.

Relation: $N1/N2=\Phi 2/\Phi 1$.

Calculation: $N2=N1\times\Phi 1/\Phi 2=1000/0.8=1250$ rpm.

Final answer: 1250 rpm.

Common mistake: Speed is inversely, not directly, proportional to flux.

Solved numerical M4-2: Armature resistance for target speed

Problem: A 220 V shunt motor runs at 1000 rpm taking Ia=30 A. Ra=0.4 Ω. Find external resistance for 700 rpm at the same current and flux.

Initial Eb: $E1=220-30\times0.4=208$ V.

At constant flux: $E2/E1=N2/N1=700/1000$; $E2=145.6$ V.

Total armature resistance: $Rtotal=(220-145.6)/30=2.48\ \Omega$.

External resistance: $Rext=2.48-0.4=2.08\ \Omega$.

Final answer: 2.08 Ω.

3. Series-motor speed control

Field diverter

A variable resistor connected in parallel with the series field diverts part of armature current, reducing field current and flux. Speed increases.

Armature diverter

A resistor is connected in parallel with armature. Series-field current can exceed armature current, increasing flux and lowering speed. It is inefficient and used for special low-speed control.

Tapped field

Selecting fewer series-field turns weakens flux and increases speed. Used in traction for discrete high-speed settings.

Series resistance

External armature resistance reduces motor voltage and speed. It is mainly used during starting and low-speed acceleration because energy is lost as heat.

4. Electric traction and traction-motor requirements

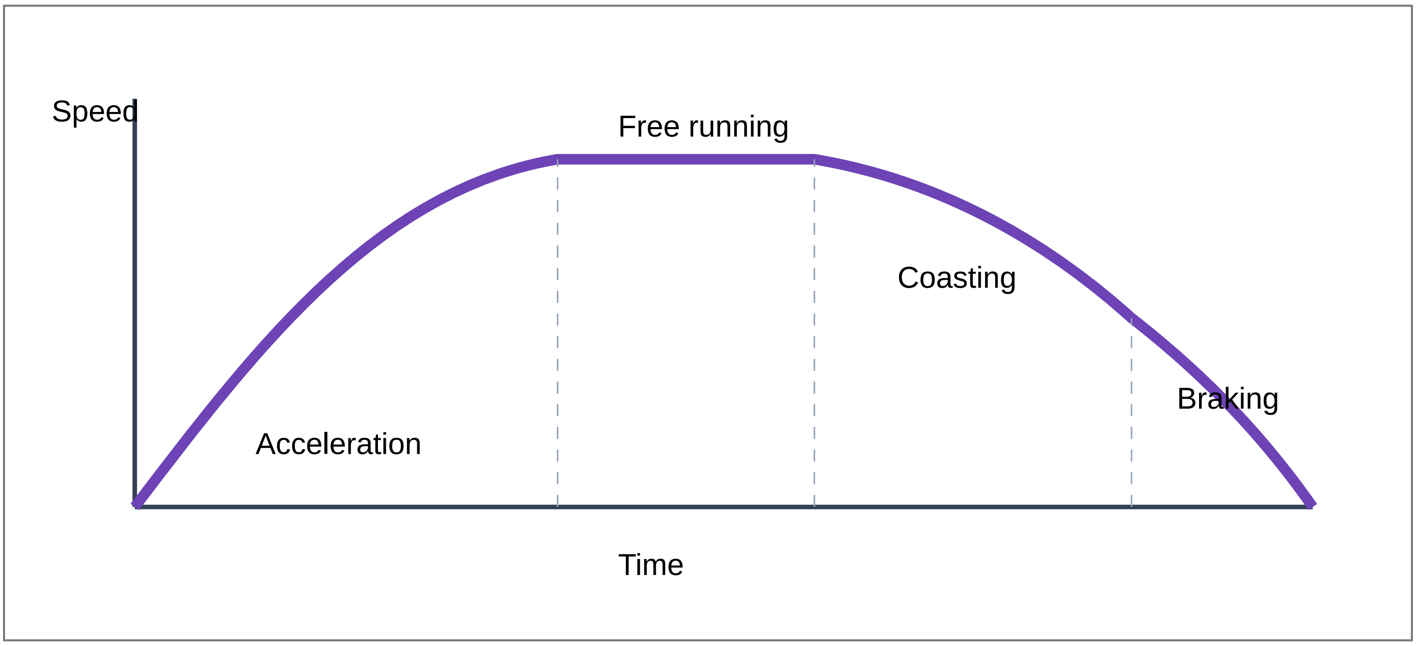
Electric traction uses electric motors to propel locomotives, multiple units, trams, trolley buses and some industrial vehicles. A traction motor repeatedly starts heavy mass, accelerates rapidly, operates over a wide speed range and must survive vibration, overload and frequent braking.

Requirement	Why it matters	DC series-motor suitability
High starting torque	Large train mass must be accelerated from rest	$T \propto Ia^2$ before saturation
High overload capacity	Gradients and acceleration require short-time overload	Robust series field and high torque
Wide speed control	Starting, acceleration, cruising and grade control	Series-parallel, resistance and field control available
Electrical braking	Reduce mechanical-brake wear and improve control	Can operate as generator for rheostatic/regenerative braking
Compact and rugged	Limited bogie space and severe vibration	Historically well suited

Malayalam Support: DC series motor-□□ starting torque □□□□ □□□□□□□□□□. Train □□□□□□□□ □□□□ load rest-□ □□□□□□ □□□□□□□□□ □□□□□□□□□□ □□□ □□□□□□□□□□□□□□.

5. Tractive effort and speed-time curve

A typical run has acceleration, free running/coasting and braking periods. The speed-time curve helps estimate schedule speed, energy use and motor rating.



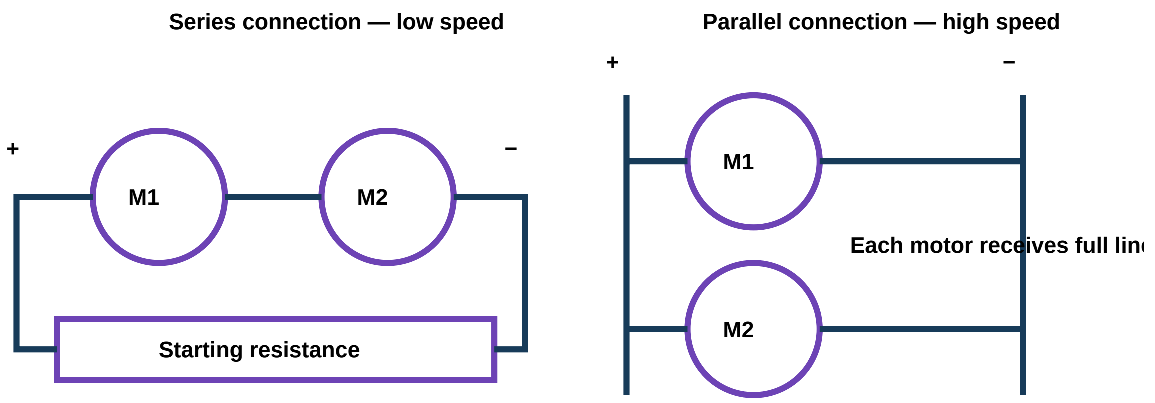
6. Series-parallel control of traction motors

For a vehicle with two identical series motors and a fixed DC supply, starting resistances limit current. At low speed, motors are connected in series so each receives about half the supply. At higher speed, motors are reconnected in parallel so each receives full supply. Resistance is progressively cut out in each connection.

Interactive traction-control sequence

Stage 1 — Series with full starting resistance

Motors M1 and M2 are in series. Current passes through all starting-resistance sections. Torque is controlled and each motor receives a fraction of supply voltage.



Current paths and switchgear are simplified for learning. Practical controllers use contactors and interlocks to prevent open circuits or short circuits during transition.

7. Transition methods

Shunt transition

During change from series to parallel, one motor is temporarily shunted by a resistor while connections are rearranged. Motor current is maintained, but some energy is lost in transition resistors.

Bridge transition

Motors and transition resistances form a bridge network. Current remains more nearly continuous and torque disturbance is lower, but switching is more complex.

Point	Shunt transition	Bridge transition
Current continuity	Maintained through shunt path	Maintained through bridge paths
Torque disturbance	Moderate	Usually lower
Control complexity	Simpler	More contactors and sequencing
Energy loss	Transition resistor loss	Transition resistor loss but smoother sharing

Malayalam Support: Series connection-□ □□□□□ motors supply voltage □□□□□□□□□□□. Parallel connection-□ □□□ motor-□□□ full voltage □□□□□□□□□□□. Transition □□□□□□□ current

□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□ shunt/bridge method.

8. Electrical braking

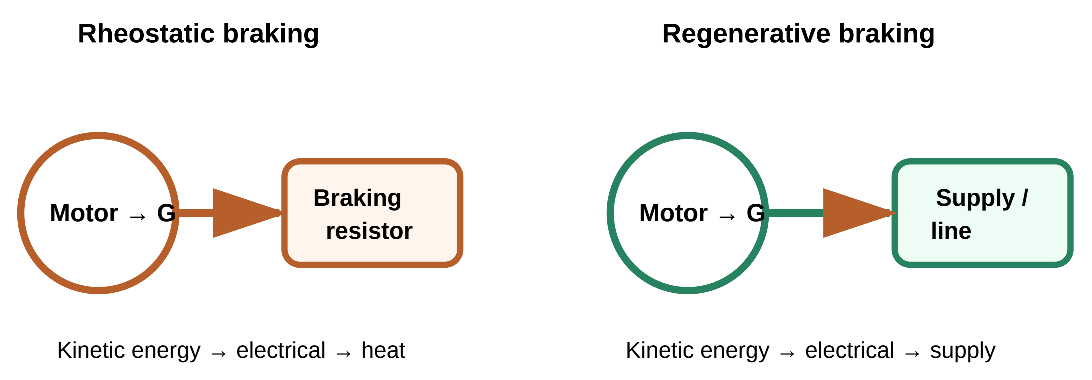
During electrical braking the motor operates as a generator and converts vehicle kinetic energy into electrical energy. The generated energy is either dissipated in resistors or returned to the supply.

Rheostatic braking

Disconnect the motor from supply, maintain suitable field and connect armature to a braking resistor. Generated current produces torque opposite motion. Energy becomes heat in the resistor.

Regenerative braking

Adjust excitation/speed so generated EMF exceeds supply voltage with correct polarity. Current reverses into the supply and electromagnetic torque opposes motion.



Arrow direction shows energy, not merely current. Mechanical brakes are still required for low speed, parking and emergency backup.

Feature	Rheostatic braking	Regenerative braking
Energy destination	Dissipated as heat	Returned to supply
Supply receptivity	Not required	Supply must accept returned energy
Control	Relatively simple	Requires correct voltage/polarity and protection
Efficiency	Energy wasted	Energy recovered
Low-speed performance	Weakens as generated EMF falls	Also weakens; mechanical brake needed

Malayalam Support: Rheostatic braking-□ generated energy resistor-□ heat □□□ □□□□□□□□□□□□□□□. Regenerative braking-□ energy supply-□□□□□□ □□□□□□□ □□□□□□□□□.

Solved numerical M4-3: Rheostatic braking power

Problem: During braking, a traction motor generates 300 V and is connected to total resistance 2.5 Ω. Neglect losses. Find braking current and electrical braking power.

Current: $I = E_g/R = 300/2.5 = 120$ A.

Power: $P = E_g I = 300 \times 120 = 36{,}000$ W.

Final answer: Braking current=**120 A**; power dissipated=**36 kW**.

Solved numerical M4-4: Regenerated power

Problem: A motor generates 620 V while connected to a 600 V line. Total circuit resistance is 0.20 Ω. Find current returned and power delivered to the line.

Current: $I = (E_g - V)/R = (620 - 600)/0.20 = 100$ A.

Line power: $P_{line} = VI = 600 \times 100 = 60$ kW.

Generated electrical power: $E_g I = 62$ kW; 2 kW is circuit copper loss.

Final answer: **100 A** returned; **60 kW** delivered to line.

Module 4 expected-question practice

Formula

State the DC-motor speed equation.

$N \propto (V - I_a R_a)/\Phi$.

Important numerical

Find speed after flux, resistance or voltage change.

Compare back EMFs and fluxes; state constant-current/flux assumptions.

Application

Why is a DC series motor suitable for traction?

High starting torque, overload capability, wide control and electrical braking.

Important diagram

Draw series and parallel motor connections.

Show starting resistances and current paths.

Comparison

Shunt transition versus bridge transition.

Compare current continuity, torque disturbance and switching complexity.

Long answer

Explain rheostatic and regenerative braking.

Include energy flow, operating condition, advantages and limitations.

FORMULA BANK

Module-wise equations, units, conditions and common mistakes

Use SI units unless a problem states otherwise. Convert mWb to Wb, kW to W and rpm to rad/s where required.

Module 1 — Generator formulas

Generated EMF

$$E_g=P\Phi ZN/(60A)$$

Units: Eg V, Φ Wb, N rpm.

Condition: Average generated DC value at brushes.

Mistake: Wrong parallel paths. Lap A=P; wave A=2.

Mini example: P=4, Φ=.02, Z=500, N=1200, wave → Eg=400 V.

Terminal voltage

$$V=E_g-I_aR_a-V_b$$

Units: V, A, Ω.

Condition: Generator delivering current.

Mistake: Brush drop must be included only if given/assumed.

Shunt current relation

$$I_{sh}=V/R_{sh};\ I_a=I_L+I_{sh}$$

Condition: Shunt generator.

Mistake: Armature current is greater than load current.

Series generator

$$I_a=I_L;\ V=E_g-I_a(R_a+R_{se})$$

Condition: Series field and armature carry same current.

Generator efficiency

$$\eta_g=P_{out}/P_{in}\times 100\%$$

Output: VIL. **Input:** output + all losses.

Mistake: Do not divide output by generated power if rotational losses exist.

Maximum efficiency

$$\text{Variable copper loss} = \text{constant loss}$$

Typical: $I^2R=W_c$.

Condition: Approximately constant voltage and constant rotational/field losses.

Module 2 — Characteristic rules

Critical resistance

$$R_{crit}=\text{slope of tangent from origin to OCC}=E/I_f$$

Maximum field-circuit resistance for build-up at a given speed. Read E and If from the tangent, not an arbitrary OCC point.

Critical speed

$$N_{crit}/N_1 \approx R_f/R_{crit,1}$$

Use only when OCCs scale proportionally with speed in the unsaturated region. Critical speed is the minimum speed for the given field resistance.

Voltage regulation

$$\%\ \text{regulation}=(V_{NL}-V_{FL})/V_{FL}\times 100\%$$

Use when asked. Keep the denominator convention stated by the problem or institution.

Commutation reactance voltage

$$e_r=L\cdot di/dt$$

Shows why rapid current reversal in an inductive coil causes sparking. This is a concept relation, not normally a detailed numerical topic here.

Module 3 — Motor formulas

Voltage and back EMF

$$V=E_b+I_aR_a;\ E_b=V-I_aR_a$$

Neglect brush drop unless stated. Back EMF is less than supply during motoring.

Back-EMF equation

$$E_b=P\Phi ZN/(60A)$$

Same machine constant as generator EMF. Use motor speed and flux.

Speed relation

$$N\propto E_b/\Phi=(V-I_aR_a)/\Phi$$

For comparisons, use ratios and state which quantities are constant.

Torque

$$T=0.159(PZ/A)\Phi I_a$$

Unit: N·m. **Mistake:** Φ must be in webers, not mWb.

Mechanical power developed

$$P_m=E_bI_a$$

This is gross converted mechanical power before rotational losses.

Torque from power

$$T=P/\omega=60P/(2\pi N)=9.55P/N$$

If P is watts and N rpm, T is N·m. With P in kW: T=9550P(kW)/N.

Starting current

$$I_{a,start}=V/(R_a+R_{st})$$

At standstill Eb=0. Solve Rst=V/Ilimit−Ra.

Motor efficiency

$$\eta_m=P_{shaft}/P_{electrical\ input}\times 100\%$$

Line input includes field current in a shunt motor.

Maximum mechanical power

$$E_b=V/2$$

Idealised armature-circuit result. It gives excessive current and ≤50% conversion efficiency before rotational losses.

Module 4 — Speed-control and braking formulas

General speed ratio

$$N_2/N_1=[(V_2-I_2R_2)/(V_1-I_1R_1)]\cdot(\Phi_1/\Phi_2)$$

R includes armature plus external series resistance. Use consistent currents.

Constant flux

$$N_1/N_2=E_{b1}/E_{b2}$$

Suitable for armature-voltage or resistance-control comparisons.

Approximately constant Eb

$$N_1/N_2=\Phi_2/\Phi_1$$

Used for field-control estimates when armature drop is small.

Rheostatic braking

$$I=E_g/R_{total};\ P=E_gI=I^2R$$

Generated energy is dissipated in resistors.

Regenerative braking

$$I=(E_g-V)/R_{total},\ \text{when } E_g>V$$

Current returns to the supply; line power approximately VI.

Mechanical output / braking torque

$$T=9.55P/N$$

Use the converted mechanical/electrical power consistent with the requested torque and account for losses if supplied.

DC MACHINE CONSTRUCTION BANK

Parts, magnetic path and maintenance insight

Yoke

Outer frame, mechanical support and flux return. Inspect for cracks, loose feet and earthing.

Pole core

Supports field coil and carries flux. Tight pole bolts preserve air-gap uniformity.

Pole shoe

Spreads flux, lowers reluctance and supports field coil.

Field winding

Creates main MMF. Shunt coils use fine wire/many turns; series coils use thick wire/few turns.

Armature core

Laminated steel cylinder with slots. Laminations reduce eddy-current loss.

Armature winding

Coils in slots connected to commutator segments. Insulation must withstand thermal and mechanical stress.

Commutator

Copper segments insulated by mica. Surface should be concentric, smooth and properly undercut.

Brushes

Carbon contacts collect/supply current. Correct grade, pressure and neutral position prevent sparking.

Shaft

Supports armature and transmits torque. Check alignment and key condition.

Bearings

Maintain rotation and air gap. Lubrication, noise and temperature indicate condition.

Interpoles

Series-connected auxiliary poles that improve current reversal under brushes.

Compensating winding

Pole-face conductors that cancel armature MMF under load.

Machine insight

Many brush and commutation faults are actually caused by mechanical problems: eccentric commutator, vibration, incorrect spring pressure, dirty surface, poor alignment or overloaded bearings.

CIRCUIT AND CONNECTION DIAGRAM BANK

What to show in every examination diagram

Diagram	Essential labels and current path	Key equation / note
Separately excited generator	Independent field supply, armature Eg/Ra, load, Ia and IL	Field current independent of load
Shunt generator	Shunt field across terminals; Ia splits into IL and Ish	Ia=IL+Ish
Series generator	Series field in load path	Ia=IL
Compound generator	Both shunt and series fields; show long/short-shunt connection	Mark aiding/opposing polarity
Shunt motor	Armature and shunt field across supply; Ia and Ish branches	IL=Ia+Ish
Series motor	Series field and armature in one path	Never run unloaded
Compound motor	Both fields; mark cumulative direction	Series field aids shunt field
Three-point starter	L,A,F; starting resistance, handle, NVR, OLR	NVR in field circuit
Four-point starter	L,A,F,N; NVR across supply	Independent of field current
Load test	Supply, starter, ammeter, voltmeter, motor, brake/dynamometer and tachometer	Pout=2πNT/60
Swinburne test	No-load shunt motor, V/A meters, field rheostat	Find constant losses from no-load input
Flux control	Field rheostat in shunt-field circuit	Weak field → high speed
Armature control	Series resistor or variable armature supply	Lower Eb → lower speed
Traction series/parallel	Two motors, starting resistors, contactors and current path	Series low speed; parallel high speed
Rheostatic braking	Motor reconnected as generator to resistor	Energy → heat
Regenerative braking	Generated EMF opposing and exceeding supply	Energy → supply

Diagram-marking rule

Use standard symbols, show terminal names, draw current arrows, state switch position and add a one-sentence working principle below the diagram.

CHARACTERISTIC CURVE BANK

Axes, shape and interpretation

Curve	Horizontal axis	Vertical axis	Important interpretation
Generator OCC	Field current If	Generated EMF Eg	Residual voltage, linear region, knee and saturation
Internal generator	Load current	Generated EMF after armature reaction	Below ideal/no-reaction curve
External generator	Load current	Terminal voltage V	Below internal curve by resistive drops
Series generator	Load current	Terminal voltage	Rises, saturates, may fall
Compound generator	Load current	Terminal voltage	Under, flat or over compounded
Shunt motor T–Ia	Armature current	Torque	Nearly straight line through origin
Shunt motor N–Ia	Armature current	Speed	Slight droop with load
Series motor T–Ia	Armature current	Torque	Quadratic before saturation; linear after
Series motor N–Ia	Armature current	Speed	Very high at light load; falls with current
Speed-control curve	Torque/load	Speed	Armature resistance gives drooping lower-speed curves; voltage control gives nearly parallel curves

Characteristic insight

A complete graph answer needs labeled axes, operating condition (constant speed/voltage/flux), important regions such as saturation, and a short physical reason for the curve shape.

POWER-FLOW AND LOSSES BANK

Where power goes in generators and motors

Generator

Mechanical input → subtract iron/mechanical losses → gross electrical power EgIa → subtract armature/field/brush losses → terminal output VIL.

Motor

Electrical input VI_L → subtract field/armature/brush losses → gross mechanical power E_bI_a → subtract rotational/stray losses → shaft output.

Loss	Cause	Variation	Reduction
Armature copper $I_a^2R_a$	Armature resistance	Variable with current squared	Low-resistance copper, cooling
Field copper	Field-winding resistance	Shunt nearly constant; series variable	Correct conductor size
Hysteresis	Repeated magnetisation of armature iron	Depends on frequency and flux density	Silicon steel
Eddy current	Induced currents in iron	Depends on flux and speed	Thin insulated laminations
Mechanical	Bearing/brush friction and windage	Approximately constant near rated speed	Lubrication, alignment and ventilation design
Stray load	Leakage, armature reaction, current redistribution	Increases with load	Good magnetic and winding design

MOTOR SELECTION AND APPLICATION BANK

Select by torque, regulation, overload and control

Application	Load requirement	Recommended motor	Technical justification
Lathe / machine tool	Nearly constant speed, adjustable range	Shunt or separately excited	Good regulation and convenient field/voltage control
Fan / blower	Low starting torque, smooth speed	Shunt	Stable speed and moderate starting duty
Centrifugal pump	Nearly constant speed	Shunt	Good regulation
Conveyor	Higher starting torque, varying load	Cumulative compound	Good starting torque with better regulation than series
Crane / hoist	Very high starting torque and overload	Series	Torque rises strongly with current
Elevator	High starting torque and controlled acceleration	Series or compound with control	Strong starting performance
Rolling mill	Shock load and high overload	Cumulative compound	Torque support with acceptable speed change
Electric traction	Very high starting torque, wide control, braking	DC series (traditional)	Excellent torque and series-parallel control
Printing machine	Controlled nearly constant speed	Shunt / separately excited	Good regulation
Reciprocating pump / press	Fluctuating heavy torque	Cumulative compound	Handles peak torque without extreme speed rise

Selection method

First identify starting torque and overload requirement, then speed regulation, duty cycle, control range, braking and supply. Do not select a motor only from power rating.

TRACTION SYSTEM BANK

Series-parallel stages, transition and braking

Stage / method	Connection or energy path	Purpose	Main limitation
Series + resistance	Motors in series with starting resistors	Limit starting current	High resistor loss
Full series	Motors in series, resistors removed	Efficient low/medium speed	Each motor receives about half voltage
Shunt transition	Temporary shunt paths during reconnection	Maintain motor current	Torque disturbance and resistor loss
Bridge transition	Bridge network balances current paths	Smoother series-to-parallel change	More complex contactor sequence
Parallel + resistance	Motors parallel with resistance	Limit current after transition	Temporary loss
Full parallel	Each motor across supply	High-speed operation	Higher voltage per motor
Rheostatic braking	Generated energy to resistor	Electrical braking independent of receptive line	Energy wasted as heat
Regenerative braking	Generated energy to supply	Energy recovery and reduced brake wear	Needs receptive supply and control

SOLVED NUMERICAL EXAMPLES

Additional worked problems across all modules

These examples complement the in-module problems and follow the required examination format.

S1. Flux per pole from generator data

Problem: A 4-pole simplex wave generator has $Z=720$, $N=500$ rpm and $E_g=240$ V. Find flux per pole.

Formula: $\Phi=60AE_g/(PZN)$, with $A=2$.

Substitution: $\Phi=(60\times2\times240)/(4\times720\times500)=0.02$ Wb.

Answer: 0.02 Wb = 20 mWb per pole.

S2. Number of armature conductors

Problem: A 6-pole lap generator must generate 250 V at 750 rpm with Φ=30 mWb. Find Z.

Formula: $Z=60AE_g/(P\Phi N)$, A=P=6.

Calculation: $Z=(60\times6\times250)/(6\times0.03\times750)=666.7$.

Answer: Use approximately **668 conductors** (an even practical integer).

S3. Lap versus wave generated voltage

Problem: A 4-pole armature has Z=480, Φ=25 mWb and N=1000 rpm. Find Eg for simplex lap and wave windings.

Lap: A=4; $E_g=(4\times.025\times480\times1000)/(60\times4)=200$ V.

Wave: A=2; $E_g=(4\times.025\times480\times1000)/(60\times2)=400$ V.

Answer: Lap=**200 V**; wave=**400 V**. The wave winding doubles series conductors per path.

S4. Shunt-generator terminal voltage and currents

Problem: Eg=250 V, Ra=0.15 Ω, Rsh=100 Ω and load current IL=80 A. Neglect brush drop. Find V, Ish and Ia.

Because $I_{sh}=V/R_{sh}$ and $I_a=I_L+I_{sh}$, $V=E_g-R_a(I_L+V/R_{sh})$.

$V[1+R_a/R_{sh}]=E_g-R_aI_L=250-12=238$.

$V=238/(1.0015)=237.64$ V; $I_{sh}=2.376$ A; $I_a=82.376$ A.

Answer: V≈**237.6 V**, Ish≈**2.38 A**, Ia≈**82.38 A**.

S5. Generator efficiency from losses

Problem: A 230 V generator supplies 60 A. Armature copper loss=500 W, field loss=250 W, iron+mechanical loss=700 W and stray loss=150 W. Find efficiency.

Output: $230\times60=13{,}800$ W. **Total loss:** 1600 W. **Input:** 15,400 W.

Efficiency: $13{,}800/15{,}400\times100=89.61\%$.

Answer: 89.6%.

S6. Voltage regulation

Problem: A shunt generator has no-load voltage 250 V and full-load voltage 230 V. Find regulation using full-load voltage as denominator.

% regulation=(250−230)/230×100=8.70%.

Answer: 8.7%.

S7. Motor speed from machine constants

Problem: A 4-pole wave motor has Z=600, Φ=20 mWb and Eb=240 V. Find speed.

Formula: N=60AEb/(PΦZ), A=2.

N=(60×2×240)/(4×.02×600)=600 rpm.

Answer: 600 rpm.

S8. Shaft torque

Problem: A motor delivers 12 kW at 900 rpm. Find shaft torque.

Formula: Tsh=9550P(kW)/N.

Tsh=9550×12/900=127.33 N·m.

Answer: 127.3 N·m.

S9. General torque equation

Problem: A 6-pole lap motor has Z=720, Φ=25 mWb and Ia=50 A. Find armature torque.

A=P=6. T=0.159(PZ/A)ΦIa=0.159×720×.025×50=143.1 N·m.

Answer: 143 N·m.

S10. Maximum-power operating point

Problem: A 240 V motor has Ra=0.4 Ω. Under ideal assumptions find Eb, Ia and developed mechanical power at maximum power.

Eb=V/2=120 V. Ia=(240−120)/0.4=300 A.

Pm=EbIa=120×300=36 kW. Armature copper loss is also 300^2×.4=36 kW.

Answer: Eb=**120 V**, Ia=**300 A**, Pm=**36 kW**. This is thermally unacceptable normal operation.

S11. New speed after voltage change

Problem: A separately excited motor runs at 1200 rpm with Eb=220 V. Flux is constant. Find speed when Eb is reduced to 165 V.

$N2/N1=Eb2/Eb1=165/220=0.75$. $N2=900$ rpm.

Answer: 900 rpm.

S12. Speed with simultaneous voltage and flux change

Problem: Initially Eb1=200 V, Φ1=30 mWb and N1=1000 rpm. Find N2 if Eb2=180 V and Φ2=24 mWb.

$N2/N1=(Eb2/Eb1)(\Phi1/\Phi2)=(180/200)(30/24)=1.125$.

Answer: 1125 rpm.

EXPECTED QUESTIONS

Module-wise examination practice patterns

These are high-priority practice patterns inferred from the official syllabus. They are not guaranteed examination questions.

Module 1

- **One word:** Mechanical rectifier, winding for high current.
- **Short answer:** Purpose of laminations, commutator and brushes.
- **Diagram:** Labeled DC machine and simple-loop generator.
- **Comparison:** Lap versus wave; shunt versus series generator.
- **Derivation:** EMF equation; maximum efficiency.
- **Numerical:** Eg, N, Φ, Z, terminal voltage and efficiency.

Module 2

- **Definition:** Armature reaction, commutation, critical resistance/speed.
- **Diagram:** MNA shift, interpoles, commutation.
- **Curve:** OCC, internal, external and compound characteristics.
- **Long answer:** Voltage build-up and conditions.
- **Application:** Select generator for a stated use.
- **Numerical:** Critical resistance, critical speed and voltage regulation.

Module 3

- **Short answer:** Back EMF and starter necessity.
- **Derivation:** Torque equation and maximum mechanical power.
- **Curve:** Shunt/series motor characteristics.
- **Diagram:** Three-point/four-point starter.
- **Testing:** Load test and Swinburne's test.
- **Numerical:** Eb, Ia, speed, torque, output, efficiency and starter resistance.

Module 4

- **Formula:** Speed relation and ratios.
- **Comparison:** Flux, armature and voltage control.
- **Application:** Why series motor is used for traction.
- **Diagram:** Series-parallel connection and transition.
- **Long answer:** Rheostatic and regenerative braking.
- **Numerical:** New speed, external resistance and braking power.

PRACTICE SECTION

Objective, diagram, curve, numerical and selection exercises

A. Fill in the blanks

1. The outer frame that completes the magnetic path is the ______.
2. In a simplex wave winding, the number of parallel paths is ______.
3. The maximum field resistance for shunt-generator voltage build-up is called ______.
4. At motor starting, back EMF is ______.
5. For a DC motor, electromagnetic torque is proportional to ______.
6. The coil that holds a three-point starter handle in RUN is the ______ release.
7. In rheostatic braking, energy is dissipated in a ______.
8. During full-parallel traction operation, each motor receives ______ supply voltage.

B. True or false

1. A wave winding is preferred for low-voltage, high-current machines.
2. Interpoles are connected in series with armature current.
3. A series motor may safely run at no load.
4. At maximum mechanical power, Eb=V/2 under simplified assumptions.
5. Field weakening increases shunt-motor speed.
6. Regenerative braking returns electrical energy to a receptive supply.

C. Multiple-choice questions

No.	Question	Options
1	Mechanical rectification is performed by	A yoke • B armature core • C commutator • D pole shoe
2	A simplex lap winding has parallel paths equal to	A 2 • B P • C Z • D N
3	Critical resistance is obtained graphically from	A external curve • B OCC tangent • C speed-torque curve • D efficiency curve
4	Before saturation, series-motor torque is approximately proportional to	A Ia • B Ia^2 • C 1/Ia • D speed
5	The starter suitable for field-weakening control without NVR release is	A two-point • B three-point • C four-point • D direct switch
6	At full series traction position, two identical motors each receive approximately	A full line voltage • B half line voltage • C zero voltage • D double line voltage
7	Electrical energy is recovered in	A plugging • B rheostatic braking • C regenerative braking • D mechanical braking
8	Swinburne’s test is mainly suitable for	A series motors • B shunt machines • C induction motors • D transformers only

D. Part-identification and connection questions

1. From the construction diagram, identify the part that spreads pole flux and supports the field coil.
2. Identify the part made of copper segments separated by mica.
3. Draw a shunt generator and mark Ia, IL and Ish.
4. Draw a three-point starter and label L, A, F, NVR and OLR.
5. Draw two traction motors in series and then in parallel, showing current path.

E. Characteristic-curve interpretation

1. Why does the external characteristic of a shunt generator fall more steeply than its internal characteristic?
2. What does the knee of an OCC indicate?
3. Why does a series-motor speed-current curve rise sharply at light current?
4. Which compound-generator curve is preferred for feeder voltage-drop compensation?

F. Formula and numerical practice

1. A 4-pole wave generator has Z=800, Φ=20 mWb and N=600 rpm. Find Eg.
2. A 6-pole lap generator has Eg=300 V, Z=900 and N=1000 rpm. Find Φ.
3. A 250 V shunt generator delivers 50 A; Ish=2 A and Ra=0.2 Ω. Find Eg, neglecting brush drop.
4. A 220 V motor has Eb=205 V and Ra=0.3 Ω. Find Ia.
5. A motor develops 10 kW at 750 rpm. Find torque.
6. A shunt motor runs at 900 rpm. Flux is reduced to 75% with unchanged Eb. Find new speed.
7. A 240 V motor with Ra=0.4 Ω must limit starting current to 60 A. Find starter resistance.
8. A braking generator produces 360 V across 3 Ω. Find current and power.

G. Motor-selection scenarios

1. Select a motor for a lathe requiring nearly constant speed and explain.

2. Select a motor for a crane lifting heavy loads from rest.
3. Select a motor for a conveyor with fluctuating load but acceptable speed regulation.
4. Select a speed-control method for efficient smooth control below base speed.
5. Select a braking method where recovered energy can be returned to the line.

H. Self-assessment checklist

- ✓ I can label all main DC-machine parts and trace the magnetic path.
- ✓ I can derive $E_g=P\Phi ZN/60A$ and choose the correct parallel paths.
- ✓ I can calculate generator voltage, losses and efficiency.
- ✓ I can explain armature reaction, commutation and voltage build-up.
- ✓ I can draw and interpret generator characteristic curves.
- ✓ I can derive motor torque and maximum-power condition.
- ✓ I can select motors from performance requirements.
- ✓ I can explain starters and Swinburne's test.
- ✓ I can solve flux, resistance and voltage speed-control problems.
- ✓ I can trace series-parallel traction stages and braking energy flow.

QUICK REVISION

Compact last-day revision sheet

Construction

Yoke, poles, field coils, laminated armature, winding, commutator, brushes, shaft, bearings, interpoles and compensating winding.

Generator principle

Flux cutting induces EMF. Right-hand rule gives direction. Commutator makes brush output unidirectional.

EMF equation

$E_g=P\Phi ZN/60A$. Lap $A=P$; wave $A=2$.

Generator voltage

$V = E_g - I_a R_a - V_b$. Shunt: $I_a = I_L + I_{sh}$.

Losses

Copper, iron, mechanical and stray. Maximum efficiency when variable copper loss equals constant loss.

Windings

Lap: high current/low voltage. Wave: high voltage/lower current.

Armature reaction

Armature MMF distorts flux and shifts MNA. Compensating winding cancels pole-face distortion.

Commutation

Current reversal under brush. Carbon brush and interpole improve commutation.

Characteristics

OCC: E_g–I_f. Internal includes armature reaction. External includes voltage drops.

Build-up

Residual magnetism, correct field connection, $R_f < R_{crit}$ and speed $> N_{crit}$.

Motor principle

Force on current-carrying conductor. Left-hand rule. Commutator keeps torque unidirectional.

Back EMF

$E_b = V - I_a R_a$. It limits current and makes the motor self-regulating.

Torque

$T=0.159(PZ/A)\Phi I_a$. $P_m=E_bI_a$. $T_{sh}=9550P(kW)/N$.

Maximum power

$E_b=V/2$ under simplified assumptions; not a normal efficient operating point.

Characteristics

Shunt: nearly constant speed. Series: very high starting torque and dangerous no-load speed.

Starters

Three-point NVR in field circuit; four-point NVR across supply. OLR protects against excess current.

Testing

Load test gives actual performance. Swinburne predicts efficiency from no-load losses.

Speed control

Flux control above base; armature resistance below base but inefficient; voltage control efficient below base.

Traction

Series motors for high starting torque. Motors series at low speed and parallel at high speed.

Braking

Rheostatic: energy to resistor. Regenerative: energy returned to supply.

Common examination mistakes

Using mWb without conversion; wrong A for lap/wave; mixing generator and motor voltage signs; using line current instead of armature current; omitting field loss; drawing curves without axes; saying a series motor can run unloaded; confusing maximum power with maximum efficiency; claiming window.print automatically downloads a PDF; omitting energy-flow arrows in braking diagrams.

Important derivations

Generator EMF equation, generator maximum-efficiency condition, motor torque equation and motor maximum-mechanical-power condition.

Important diagrams

DC machine, simple-loop generator, armature reaction, commutation, OCC/critical tangent, three-point and four-point starters, motor characteristics, series-parallel traction and both electrical-braking circuits.

FRESH MODEL QUESTION PAPER

DC Machines & Traction Motors — Course 3032

Time: 3 Hours | Maximum Marks: 75

PART A

Answer all questions. Each carries 1 mark. (9 × 1 = 9)

1. Name the part of a DC machine that acts as a mechanical rectifier.
2. Write the number of parallel paths in a simplex wave winding.
3. State one essential condition for voltage build-up of a shunt generator.
4. Name the auxiliary pole used to improve commutation.
5. Write the motor voltage equation, neglecting brush drop.
6. Which DC motor has the highest starting torque?
7. Name the release that protects a starter against excessive armature current.
8. Which speed-control method is normally used above base speed in a shunt motor?
9. Name the braking method that returns energy to the supply.

PART B

Answer any eight questions. Each carries 3 marks. (8 × 3 = 24)

1. State the functions of yoke, pole shoe and commutator.
2. Compare simplex lap and simplex wave windings.
3. A 4-pole wave generator has 600 conductors, flux 20 mWb/pole and speed 1000 rpm. Calculate generated EMF.
4. Define armature reaction and list two effects.
5. Distinguish critical field resistance and critical speed.
6. Explain why back EMF makes a DC motor self-regulating.
7. Write the torque equation and define each symbol.
8. Compare three-point and four-point starters.
9. State the procedure of Swinburne's test.
10. Compare rheostatic and regenerative braking.

PART C

Answer one alternative from each group. Each group carries 7 marks. (6 × 7 = 42)

1. **(A)** Explain the construction and working principle of a DC generator with a labeled diagram and derive the EMF equation. **OR (B)** A 6-pole lap generator has Z=720, Φ=30 mWb, N=800 rpm, Ra=0.2 Ω and supplies 100 A. Field current is 2 A. Find Eg, Ia, terminal voltage and output power. Neglect brush drop.
2. **(A)** Classify DC-generator losses, draw the power-flow diagram and derive the maximum-efficiency condition. **OR (B)** Explain armature windings and compare lap and wave windings with applications.
3. **(A)** Explain armature reaction, interpoles and compensating windings with diagrams. **OR (B)** Explain shunt-generator OCC, critical resistance, critical speed and voltage build-up conditions.
4. **(A)** Derive the general torque equation of a DC motor and explain shunt and series motor characteristics. **OR (B)** A 220 V shunt motor takes line current 45 A; Ish=1.5 A, Ra=0.25 Ω and speed=950 rpm. Rotational losses are 600 W. Find Eb, developed torque, shaft output and efficiency.
5. **(A)** Explain construction and working of three-point and four-point starters. **OR (B)** Explain the direct load test and Swinburne's test, including advantages and limitations.

6. **(A)** Explain shunt- and series-motor speed-control methods with diagrams and comparison. **OR** **(B)** Explain series-parallel traction control, shunt/bridge transition, rheostatic braking and regenerative braking.

COMPLETE MODEL ANSWERS

Full solutions for all questions and both OR alternatives

Part A

No.	Answer
1	Commutator
2	2
3	Any one: residual magnetism, correct field polarity, field resistance below critical value, speed above critical speed
4	Interpole / commutating pole
5	V=Eb+IaRa
6	DC series motor
7	Overload release
8	Flux or field-control method
9	Regenerative braking

Part B

B1. Functions

Yoke: provides mechanical support and a low-reluctance return path for flux. **Pole shoe:** spreads flux over the armature, reduces air-gap reluctance and supports the field coil. **Commutator:** acts as a mechanical rectifier in a generator and reverses armature-coil current at the correct instant in a motor.

B2. Lap versus wave

Simplex lap has A=P, usually P brush sets, is suited to low-voltage high-current duty and needs equalizer rings. Simplex wave has A=2, usually two brush sets, is suited to high-voltage lower-current duty and normally does not need equalizer rings.

B3. Generated EMF

P=4, A=2, Z=600, Φ=.02 Wb, N=1000 rpm.

Eg=PΦZN/(60A)=(4×.02×600×1000)/(60×2)=400 V.

Answer: 400 V.

B4. Armature reaction

Armature reaction is the effect of armature-current MMF on the main field. It distorts flux, shifts MNA, reduces average flux/generated voltage and can cause brush sparking and local pole-tip heating.

B5. Critical resistance and speed

Critical resistance is the maximum shunt-field resistance at which a generator just builds voltage at a specified speed; graphically it is the tangent slope from the OCC origin. Critical speed is the minimum speed at which voltage builds for a given field resistance.

B6. Self-regulation by back EMF

When load rises, speed and Eb fall slightly. Since $I_a=(V-E_b)/R_a$, armature current and torque rise to meet the load. When load falls, speed/Eb rise, current falls and torque reduces.

B7. Torque equation

$T=0.159(PZ/A)\Phi I_a$ N·m, where P is poles, Z total armature conductors, A parallel paths, Φ flux per pole in Wb and Ia armature current in A.

B8. Three-point versus four-point starter

Three-point terminals are L,A,F and its NVR is in series with the shunt field; field weakening may release the handle, but field failure is sensed. Four-point terminals are L,A,F,N and its NVR is connected across the supply independently, so field weakening does not release it; separate field-failure protection is desirable.

B9. Swinburne’s procedure

Run the shunt motor at rated voltage without mechanical load. Measure V, no-load line current I0, field current Ish and Ra. Find $I_{a0}=I_0-I_{sh}$. Constant rotational loss is $VI_0-VI_{sh}-I_{a0}^2R_a$, with brush loss included if specified. Use this constant loss plus copper losses at an assumed load to predict efficiency.

B10. Braking comparison

In rheostatic braking the motor generates into a resistor and energy becomes heat; it does not require a receptive supply but wastes energy. In regenerative braking Eg exceeds supply voltage with correct polarity, current returns to the supply and energy is recovered; more control and a receptive supply are required.

Part C — Group 1

C1(A). Construction, principle and EMF derivation

Construction: The yoke supports the machine and carries return flux. Main poles and field windings create stationary flux. The laminated armature core carries conductors and rotates on the shaft. Copper commutator segments insulated by mica connect armature coils to stationary carbon brushes. Bearings support rotation; interpoles and compensating windings improve loaded operation.

Working principle: A prime mover rotates armature conductors through magnetic flux. By Faraday's law an EMF is induced. Fleming's right-hand rule gives direction. The EMF in each coil reverses every half-turn, while the commutator reverses coil connection to the brushes, producing unidirectional terminal voltage.

Derivation: Flux cut by one conductor per revolution=PΦ Wb. Time per revolution=60/N s. EMF per conductor=PΦN/60 V. Conductors in series per path=Z/A. Therefore:

$$E_g=(P\Phi N/60)(Z/A)=P\Phi ZN/60A$$

For simplex lap A=P; for simplex wave A=2. A complete diagram should label yoke, pole, field coil, armature, commutator, brushes and shaft.

C1(B). Generator numerical

Given: P=6, lap A=6, Z=720, Φ=.03 Wb, N=800 rpm, Ra=.2 Ω, IL=100 A, Ish=2 A.

Generated EMF: Eg=(6×.03×720×800)/(60×6)=288 V.

Armature current: Ia=IL+Ish=102 A.

Terminal voltage: V=Eg−IaRa=288−102×.2=267.6 V.

Output power: Pout=VIL=267.6×100=26,760 W=26.76 kW.

Final: Eg=**288 V**, Ia=**102 A**, V=**267.6 V**, output=**26.76 kW**.

Part C — Group 2

C2(A). Losses, power flow and maximum efficiency

Copper losses: $I_a^2R_a$, series-field I^2R, shunt-field V^2/R_{sh} and brush-contact loss. **Iron losses:** hysteresis and eddy currents. **Mechanical losses:** bearing/brush friction and windage. **Stray-load losses:** additional loaded effects.

Power flow: Mechanical input → subtract iron and mechanical losses → gross electrical power EgIa → subtract armature, brush and field losses → terminal output VIL.

Let output=VI, variable loss=I^2R and constant loss=Wc. η=VI/(VI+I^2R+Wc). Differentiating with respect to I and setting dη/dI=0 gives I^2R=Wc.

> Maximum efficiency occurs when variable copper loss equals constant loss.

Assume nearly constant voltage and constant rotational/field losses.

C2(B). Armature windings

An armature winding consists of coils placed in slots and connected to commutator segments. Each turn has two active coil sides. Parallel paths determine current and voltage rating.

Lap winding: Coils connect to nearby commutator segments and lap back. Simplex A=P. It carries high current at lower voltage, usually uses P brush sets and needs equalizer rings.

Wave winding: Coils progress around the armature in a wave, connecting conductors under similar pole polarities. Simplex A=2. It gives higher voltage at lower current, uses two main brush sets and generally needs no equalizer rings.

Applications: Lap—heavy-current generators, electroplating and welding supplies. Wave—higher-voltage lighting or general DC generators.

Part C — Group 3

C3(A). Armature reaction, interpoles and compensation

Armature current produces armature MMF. Its field combines with the main field, crowds flux at one pole tip, weakens it at the other, shifts MNA and can reduce average flux. Effects include sparking, voltage reduction, local heating and poor commutation.

Interpoles: Small poles between main poles, series-connected with armature. Their commutating EMF opposes reactance voltage and allows brushes to remain near geometrical neutral.

Compensating winding: Conductors embedded in pole shoes and series-connected with armature. Their MMF opposes armature MMF under the pole arc, preventing flux distortion during rapid load change.

A complete diagram must show main poles, armature, shifted MNA, interpoles and pole-face compensation conductors.

C3(B). OCC, critical values and voltage build-up

The OCC plots Eg against If at constant speed and no load. It begins at residual voltage, rises approximately linearly and bends at magnetic saturation.

Critical resistance is the maximum field resistance for build-up at a given speed and equals the slope of the tangent from origin to OCC. Critical speed is the minimum speed for build-up at a fixed field resistance.

Build-up sequence: residual magnetism creates initial Eg → field current flows → field MMF aids residual flux → Eg rises → operation settles at the field-line/OCC intersection.

Conditions: residual magnetism, correct field polarity, closed field circuit, Rf below critical resistance and speed above critical speed.

Part C — Group 4

C4(A). Torque derivation and characteristics

$P_m = E_b I_a$ and $E_b = P\Phi ZN/(60A)$. Angular speed $\omega = 2\pi N/60$. Therefore:

$$T = P_m/\omega = [P\Phi ZN\, I_a/(60A)]/[2\pi N/60] = PZ\Phi I_a/(2\pi A) = 0.159(PZ/A)\Phi I_a$$

Thus $T \propto \Phi I_a$.

Shunt motor: Φ approximately constant, so $T \propto I_a$. Speed is nearly constant and drops slightly with load. Suitable for lathes, fans and pumps.

Series motor: Before saturation $\Phi \propto I_a$, so $T \propto I_a^2$; after saturation $T \propto I_a$. Speed is inversely related to flux and becomes dangerously high at light load. Suitable for cranes, hoists and traction.

C4(B). Motor numerical

Given: V=220 V, IL=45 A, Ish=1.5 A, Ra=.25 Ω, N=950 rpm, rotational loss=600 W.

Armature current: $I_a = 45 - 1.5 = 43.5$ A.

Back EMF: $E_b = 220 - 43.5 \times .25 = 209.125$ V.

Developed power: $P_m = E_b I_a = 209.125 \times 43.5 = 9096.94$ W.

Developed torque: $T_a = 9550 \times 9.09694/950 = 91.44$ N·m.

Shaft output: $9096.94 - 600 = 8496.94$ W.

Input: $220 \times 45 = 9900$ W.

Efficiency: $8496.94/9900 \times 100 = 85.83\%$.

Final: $E_b \approx$ **209.1 V**, $T_a \approx$ **91.4 N·m**, output≈**8.50 kW**, efficiency≈**85.8%**.

Part C — Group 5

C5(A). Three-point and four-point starters

A starter inserts resistance in series with armature at starting because $E_b = 0$. The handle moves over studs to cut out resistance as speed rises.

Three-point: Terminals L,A,F. OLR is in armature path and trips on excess current. NVR is in series with the shunt field and holds the handle at RUN. Supply failure or field opening releases it. Field weakening may reduce NVR current and cause unwanted release.

Four-point: Terminals L,A,F,N. NVR is connected across supply independently, so field weakening does not release the handle. It needs separate field-failure protection. Diagrams must show resistance studs, handle, spring, NVR and OLR.

C5(B). Load test and Swinburne's test

Direct load test: Couple motor to brake/dynamometer, start safely, increase load, record V, I, N and torque. Calculate input VIL, output $2\pi NT/60$ and efficiency. It gives actual temperature, commutation and efficiency but wastes full-load power and needs loading equipment.

Swinburne's test: Run shunt motor no-load at rated voltage. Measure V, I_0, I_{sh} and R_a. Determine constant rotational loss from no-load input after subtracting field and no-load armature copper loss. Predict efficiency at any load by adding load copper losses.

Advantages: economical, simple, low power, predicts motor/generator efficiency. **Limitations:** no full-load temperature/commutation test, inaccurate stray-load effects, not suitable for series motors.

Part C — Group 6

C6(A). Speed-control methods

From $N \propto (V - I_a R_a)/\Phi$:

Shunt field control: increase field resistance → reduce If and Φ → speed above base. Efficient but torque capacity falls.

Shunt armature resistance: add series resistance → larger IaR drop → lower speed. Simple but inefficient and load-dependent.

Voltage control: vary armature voltage with constant field → smooth efficient speed control below base.

Series motor: field diverter weakens field and raises speed; armature diverter strengthens effective field and lowers speed; tapped-field control changes turns; series resistance gives low-speed starting control.

Diagrams should show the rheostat/diverter position and current paths. Flux control is generally above base; armature voltage/resistance is below base.

C6(B). Traction control and braking

At starting, two series motors are connected in series with maximum starting resistance. As speed and back EMF rise, resistance is cut out to full-series. The motors are then transitioned to parallel, first with limiting resistance and finally full-parallel.

Shunt transition: temporary shunt resistors maintain current while reconnection occurs. **Bridge transition:** a bridge network gives smoother current/torque continuity but uses more complex contactors.

Rheostatic braking: motors generate into resistors; kinetic energy becomes heat. **Regenerative braking:** generated EMF exceeds line voltage with correct polarity, returning current and energy to the supply. Mechanical brakes remain necessary for low speed, parking and emergency.

ANSWER KEY

Practice answers and final numerical values

A. Fill in the blanks

1 Yoke. 2 Two. 3 Critical field resistance. 4 Zero. 5 ΦI_a. 6 No-volt. 7 Braking resistor/rheostat. 8 Full.

B. True or false

1 False—lap winding is preferred for low-voltage high-current duty. 2 True. 3 False—series motor no-load speed may become dangerous. 4 True. 5 True. 6 True.

C. MCQs

1-C, 2-B, 3-B, 4-B, 5-C, 6-B, 7-C, 8-B.

D. Part-identification and connection answers

1. Pole shoe.
2. Commutator.
3. In a shunt generator, the field is across terminal voltage; Ia reaches the brush junction and divides into IL and Ish, so $I_a = I_L + I_{sh}$.
4. The three-point starter has L to starting resistance/armature through OLR, F through the NVR/field circuit and a spring-return handle.
5. Series: supply → resistance → M1 → M2 → return. Parallel: both M1 and M2 form separate branches across the same supply.

E. Characteristic-curve answers

1. External voltage includes both armature-reaction reduction and internal resistive/brush drops; falling shunt voltage also weakens field current.
2. The knee marks onset of magnetic saturation; beyond it, large field-current increase gives small voltage increase.
3. At light current, series-field flux is small. Since $N \propto E_b/\Phi$, speed rises sharply.
4. Over-compounded cumulative generator, because its rising terminal voltage can compensate feeder drop.

F. Numerical answers

No.	Working summary	Final answer
1	$E_g=4\times.02\times800\times600/(60\times2)$	320 V
2	$\Phi=60\times6\times300/(6\times900\times1000)$	0.02 Wb = 20 mWb
3	$I_a=50+2=52$ A; $E_g=250+52\times.2$	260.4 V
4	$I_a=(220-205)/.3$	50 A
5	$T=9550\times10/750$	127.3 N·m
6	$N_2=900/0.75$	1200 rpm
7	$R_{st}=240/60-.4$	3.6 Ω
8	$I=360/3$; $P=360\times120$	120 A, 43.2 kW

G. Motor-selection answers

1. **Lathe:** shunt or separately excited motor because speed regulation is good and speed can be controlled smoothly.
2. **Crane:** series motor because very high starting torque and overload capacity are required.
3. **Conveyor:** cumulative compound motor because it combines high starting torque with better speed regulation than a series motor.
4. **Efficient below-base control:** armature-voltage control with constant field.
5. **Energy recovery:** regenerative braking, provided the supply can accept returned power.

REFERENCES AND ONLINE RESOURCES

Official syllabus references

Textbooks and reference books

Code	Book
T1	B. L. Theraja and A. K. Theraja, *Electrical Technology*, Volume II, S. Chand & Co.
T2	B. L. Theraja and A. K. Theraja, *Electrical Technology*, Volume III, S. Chand & Co.
R1	P. S. Bimbhra, *Electrical Machinery*, Khanna Publishers.
R2	N. V. Suryanarayana, *Utilization of Electric Power: Including Electric Drives and Electric Traction*, New Age International Publishers.
R3	K. Murugesh Kumar, *DC Machines and Transformers*, S. Chand & Company.
R4	J. B. Gupta, *Theory and Performance of Electrical Machines*, S. K. Kataria & Sons.
R5	M. V. Deshpande, *Electrical Machines*, Prentice Hall India, New Delhi, 2011.
R6	I. J. Nagrath and D. P. Kothari, *Electrical Machines*, Tata McGraw-Hill Education.
R7	B. R. Gupta, *Fundamentals of Electric Machines*, Third Edition, New Age International Publishers.

Official online resources listed in the syllabus

- **NPTEL 108102146** ↗
- **NPTEL 108105155** ↗
- **NPTEL 108106071** ↗
- **NPTEL 108105053** ↗
- **SWAYAM** ↗
- **COEP Electrical Machines Virtual Lab** ↗
- **IIT Guwahati Virtual Electrical Machines Lab** ↗
- **SITTTR model question-paper page for course 3032** ↗

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FINAL EXAMINATION CHECKLIST

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Before submitting the answer book

Numerical answers

- ✓ Write given data with units.
- ✓ Convert mWb to Wb and kW to W where needed.
- ✓ Identify lap/wave and correct A.
- ✓ Use Ia rather than IL where appropriate.
- ✓ State brush-drop and loss assumptions.
- ✓ Box the final answer with unit.

Theory and diagrams

- ✓ Start with definition/principle.
- ✓ Draw labeled diagram with arrows and current path.
- ✓ Explain construction and working in sequence.
- ✓ Label graph axes and operating regions.
- ✓ Give advantages, limitations and applications where asked.
- ✓ End with the required conclusion or comparison.

Seven-mark answer structure

Definition/principle → labeled diagram → construction/connection → step-by-step working → formula/curve → advantages/limitations → application or conclusion.